

DNA Replication in Prokaryotes and Eukaryotes

A Complete Molecular Biology Teaching and Learning Module

Complete Tutorial

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Module 01: Introduction to DNA Replication

Learning Objectives

By the end of this module, you will be able to:

1. Explain why DNA replication is essential for life
2. Describe the historical development of our understanding of DNA replication
3. Identify the key questions that drove replication research
4. Understand the scope of this course

Why DNA Replication Matters

Concept

DNA replication is the biological process by which a cell duplicates its entire genome prior to cell division, ensuring that each daughter cell receives a complete and accurate copy of the genetic information.

Why It Matters

Every time a cell divides — whether it is a bacterium dividing every 20 minutes or a human cell renewing tissue — the complete genome must be copied. Errors in this process can lead to:

- **Cell death** if essential genes are damaged
- **Cancer** if tumor suppressor genes or oncogenes are mutated
- **Genetic diseases** if mutations are passed to offspring
- **Evolution** if mutations are heritable and provide selective advantage

DNA replication is therefore one of the most fundamental and highly regulated processes in biology.

Beginner Explanation

Think of DNA replication as photocopying a very important book. The cell needs to make an exact copy of its "instruction manual" (DNA) so that when it divides, each new cell gets its own complete set of instructions. The cell has evolved incredibly sophisticated molecular machinery to do this with remarkable speed and accuracy.

Molecular-Level Explanation

DNA replication involves:

- **Unwinding** the double helix to expose single-stranded templates
- **Synthesizing** new complementary strands using DNA polymerase enzymes
- **Reading** the template in the 3'→5' direction while synthesizing in the 5'→3' direction
- **Proofreading** newly synthesized DNA to correct errors

- **Coordinating** thousands of proteins working simultaneously across the genome

In *E. coli*, the entire 4.6 million base pair genome is replicated in approximately 40 minutes with an error rate of roughly one mistake per 10^9 nucleotides incorporated.

Advanced Explanation

The fidelity of DNA replication is achieved through three layers:

1. **Base selection** by DNA polymerase (error rate $\sim 10^{-5}$)
2. **Proofreading** by 3'→5' exonuclease activity (improves to $\sim 10^{-7}$)
3. **Post-replicative mismatch repair** (final error rate $\sim 10^{-9}$ to 10^{-10})

This means roughly one error per billion base pairs — an extraordinary level of accuracy considering the speed of replication (*E. coli*: $\sim 1,000$ nucleotides/second; human cells: ~ 50 nucleotides/second per fork).

Historical Context

The Discovery of DNA Structure (1953)

Watson and Crick's elucidation of the double-helical structure of DNA immediately suggested a mechanism for replication. In their landmark 1953 paper, they wrote: *"It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material."*

The key insight: because A pairs with T and G pairs with C, each strand contains sufficient information to specify the complementary strand.

Three Models of Replication

After the discovery of DNA structure, three models were proposed:

1. Conservative Model

- The original double helix remains intact
- An entirely new double helix is synthesized
- The parental molecule is "conserved"

2. Semiconservative Model

- Each strand of the parental molecule serves as a template
- Each daughter molecule contains one old strand and one new strand
- Proposed by Watson and Crick

3. Dispersive Model

- The parental molecule is broken into fragments
- New and old DNA are interspersed within each strand
- Each strand is a mosaic of old and new DNA

The Meselson-Stahl Experiment (1958)

The most beautiful experiment in biology.

Matthew Meselson and Franklin Stahl provided definitive evidence for semiconservative replication using:

- *E. coli* grown in heavy nitrogen (^{15}N) medium
- Transfer to light nitrogen (^{14}N) medium
- Density gradient centrifugation (CsCl) to separate DNA by density

Results:

Generation	Expected Pattern (Semiconservative)	Observed
Parent (G0)	One heavy band ($^{15}\text{N}/^{15}\text{N}$)	✓ One heavy band
First (G1)	One hybrid band ($^{15}\text{N}/^{14}\text{N}$)	✓ One hybrid band
Second (G2)	One hybrid + one light band	✓ One hybrid + one light band
Third (G3)	One hybrid + two light bands	✓ One hybrid + two light bands

This elegant experiment conclusively demonstrated semiconservative replication.

Visual Explanation

Meselson-Stahl Experiment:

=====

Step 1: Grow *E. coli* in ^{15}N (heavy) medium

→ All DNA is heavy ($^{15}\text{N}/^{15}\text{N}$)

→ CsCl gradient: Single band at bottom

Step 2: Transfer to ^{14}N (light) medium

Generation 1 (one round of replication):

→ All molecules are hybrid ($^{15}\text{N}/^{14}\text{N}$)

→ CsCl gradient: Single band at middle position

Generation 2 (two rounds):

→ 50% hybrid ($^{15}\text{N}/^{14}\text{N}$) + 50% light ($^{14}\text{N}/^{14}\text{N}$)

→ CsCl gradient: Two bands (middle + top)

Generation 3:

→ 25% hybrid + 75% light

→ CsCl gradient: Two bands (middle + top, top band thicker)

Key Questions in Replication Biology

The field of DNA replication addresses several fundamental questions:

1. **Where** does replication start? (Origins of replication)
2. **When** does replication occur? (Cell-cycle regulation)
3. **How** is the double helix unwound? (Helicases)
4. **How** is synthesis initiated? (Primers and primase)
5. **Which** enzymes synthesize DNA? (DNA polymerases)
6. **How** is accuracy achieved? (Proofreading and repair)
7. **How** are the ends of linear chromosomes replicated? (Telomeres)
8. **What** happens when replication goes wrong? (Checkpoints and repair)

Common Misconceptions

Misconception	Reality
DNA replication copies the entire genome at once	Replication begins at specific origins and proceeds bidirectionally
Both strands are synthesized in the same direction	Leading strand: continuous 5'→3'; Lagging strand: discontinuous 5'→3'
DNA polymerase can start synthesis from scratch	DNA polymerase requires a pre-existing 3'-OH group (primer)
Replication is 100% accurate	Error rate is $\sim 10^{-9}$ – 10^{-10} after all correction mechanisms
Prokaryotic and eukaryotic replication are essentially the same	They share fundamental principles but differ in many mechanistic details

Exam Point

The Meselson-Stahl experiment is one of the most commonly examined topics. Know the three models, the experimental approach, and the predicted vs. observed results for each generation.

Quick Check

1. What are the three models of DNA replication that were proposed after the discovery of DNA structure?
2. Why does the semiconservative model predict a hybrid band after one round of replication in the Meselson-Stahl experiment?
3. If DNA polymerase can only synthesize in the 5'→3' direction, what challenge does this create for replicating the two strands of the double helix?
4. Calculate: If *E. coli* replicates its 4.6×10^6 bp genome in 40 minutes with two replication forks, what is the rate of nucleotide incorporation per fork?
5. Why is the error rate of DNA replication relevant to human health?

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Module 02: Core Concepts of DNA Replication

Learning Objectives

1. Explain the semiconservative model of replication at the molecular level
2. Describe all requirements for DNA replication
3. Explain the concept of replication origins
4. Describe replication forks, leading strand, and lagging strand
5. Explain why DNA synthesis proceeds 5'→3' and its consequences
6. Understand the overall workflow of DNA replication

1. Semiconservative Replication

Concept

Semiconservative replication means that each strand of the parental DNA double helix serves as a template for the synthesis of a new complementary strand. After replication, each daughter DNA molecule consists of one parental (original) strand and one newly synthesized strand.

Why It Matters

This mechanism ensures:

- **Faithful transmission** of genetic information
- **Each daughter cell** receives one copy of every gene
- **Template-directed synthesis** provides a mechanism for accuracy

Beginner Explanation

Imagine you have a zipper (DNA double helix). You unzip it, and each half-zipper serves as a guide to build a new matching half. The result: two identical zippers, each with one old half and one new half.

Molecular-Level Explanation

Before Replication:

5'-A-T-G-C-C-A-T-G-3' (Parental Strand 1)

3'-T-A-C-G-G-T-A-C-5' (Parental Strand 2)

↓ Unwinding + Synthesis

After Replication:

5'-A-T-G-C-C-A-T-G-3' (Parental Strand 1)

3'-T-A-C-G-G-T-A-C-5' (New Strand 1)

5'-A-T-G-C-C-A-T-G-3' (New Strand 2)

3'-T-A-C-G-G-T-A-C-5' (Parental Strand 2)

Each daughter molecule = 1 old strand + 1 new strand

Advanced Explanation

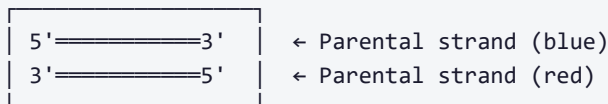
The semiconservative model is not just about copying — it has implications for:

- **Mutation tracking:** Mutations accumulate differently on leading vs. lagging strands
- **Epigenetic inheritance:** Parental histone marks are distributed to daughter chromosomes
- **Strand-specific repair:** Cells can distinguish the template strand from the new strand

Visual Explanation

SEMICONSERVATIVE REPLICATION

Parent DNA



↓
Unwinding
↓



Result: Two identical daughter molecules
Each = 1 parental strand + 1 new strand

Common Misconception

Misconception: Semiconservative means each strand is "half new and half old." **Reality:** Each *molecule* contains one complete old strand and one complete new strand. The strands themselves are entirely old or entirely new.

2. Requirements for DNA Replication

Concept

DNA replication requires specific substrates, enzymes, energy, and accessory factors working together in a coordinated molecular machine called the **replisome**.

Complete List of Requirements

A. Substrates (Building Blocks)

Component	Role	Details
dNTPs (dATP, dTTP, dGTP, dCTP)	Nucleotide substrates	Provide both building blocks and energy (pyrophosphate release drives the reaction)
Template DNA	Template for synthesis	Each strand serves as a template
Primer (RNA or DNA)	Provides free 3'-OH	DNA polymerase cannot start synthesis de novo

B. Enzymes

Enzyme	Function
DNA polymerase	Synthesizes new DNA strands (5'→3')
Helicase	Unwinds the double helix
Primase	Synthesizes RNA primers
Topoisomerase	Relieves torsional stress ahead of the fork
DNA ligase	Joins Okazaki fragments (seals nicks)
Exonuclease	Removes misincorporated nucleotides (proofreading) and RNA primers

C. Accessory Proteins

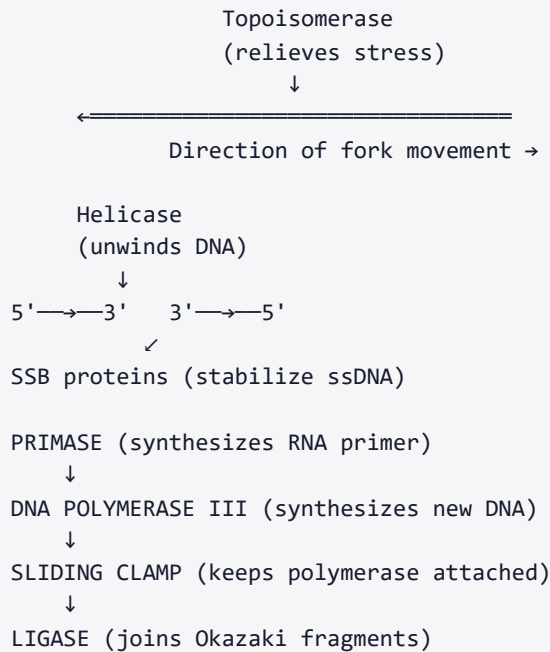
Protein	Function
Sliding clamp	Increases polymerase processivity
Clamp loader	Loads sliding clamp onto DNA
Single-strand binding protein (SSB/RPA)	Stabilizes unwound single-stranded DNA
Origin recognition proteins	Identify and bind replication origins

D. Energy

- **ATP** — used for helicase activity, clamp loading, topoisomerase action
- **Pyrophosphate release** from dNTPs — thermodynamically drives polymerization

Visual Explanation

REPLICATION FORK – All Components



3. Origin of Replication (ori)

Concept

An **origin of replication** is a specific DNA sequence where replication begins. It is the site where the replication machinery assembles and initiates DNA synthesis.

Why It Matters

Origins determine:

- **When** replication starts (regulated by cell-cycle signals)
- **Where** replication starts (ensures entire genome is copied)
- **How many** replication bubbles form (determines replication speed)

Beginner Explanation

Think of the origin of replication as the "start button" for copying DNA. Just as a photocopier needs a starting point, the cell's replication machinery needs a specific place on the DNA to begin making copies.

Molecular-Level Explanation

Prokaryotic Origins

E. coli has a **single origin** called **oriC**:

- **Length:** 245 base pairs
- **Key elements:**
 - **5 × 9-mer repeats (DnaA boxes):** Binding sites for initiator protein DnaA

- **AT-rich 13-mer repeats:** Easily unwound (A-T pairs have only 2 hydrogen bonds)
- Region melts first → creates replication bubble

Eukaryotic Origins

Human cells have **thousands of origins** (~30,000–50,000 per cell):

- **ARS (Autonomously Replicating Sequences)** in yeast: ~100–200 bp
- **ORC binding sites** in humans: less sequence-specific, influenced by chromatin context
- Each origin fires once per cell cycle (ensuring no re-replication)

Advanced Explanation

Origin specification involves:

1. **Sequence elements:** Specific DNA sequences that attract initiator proteins
2. **Chromatin context:** Epigenetic marks (histone modifications, DNA methylation) influence origin usage
3. **Transcriptional activity:** Active genes often have nearby efficient origins
4. **Replication timing domains:** Origins in euchromatin fire early; heterochromatin origins fire late

Common Misconception

Misconception: All origins are the same sequence. **Reality:** Origin sequences vary significantly between species and even within genomes. In metazoans, origins are less sequence-dependent and more influenced by chromatin structure and epigenetic factors.

4. Replication Fork

Concept

When DNA replication initiates at an origin, the double helix is unwound to form a **replication bubble**. At each end of the bubble, a Y-shaped structure called a **replication fork** forms where the actual synthesis occurs.

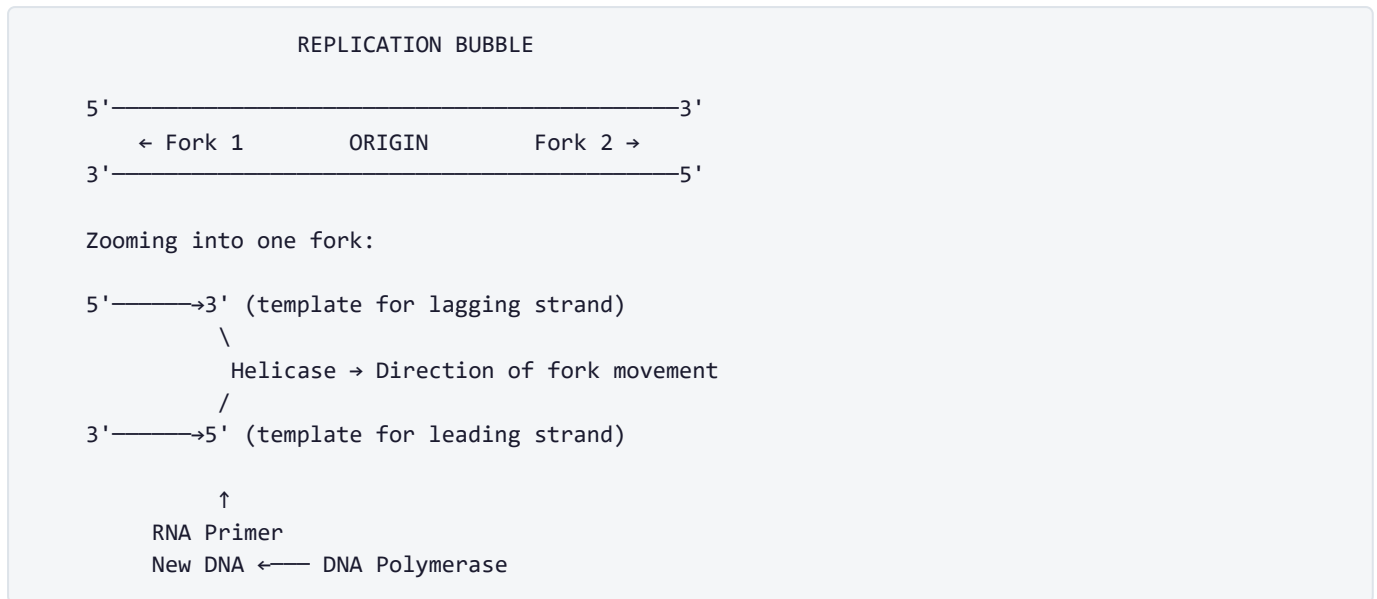
Why It Matters

The replication fork is the active site of DNA synthesis. Understanding its structure is essential for understanding how both strands are replicated simultaneously.

Beginner Explanation

Imagine unzipping a zipper from the middle. You get two openings — one on each side. Each opening is a "replication fork." At each fork, the two strands of DNA are separated, and new strands are built on each template.

Molecular-Level Explanation



Advanced Explanation

Replication forks are **asymmetric**:

- The two template strands run antiparallel ($5' \rightarrow 3'$ and $3' \rightarrow 5'$)
- DNA polymerase can only synthesize in the $5' \rightarrow 3'$ direction
- Therefore, one strand is synthesized **continuously** (leading strand) and the other **discontinuously** (lagging strand)
- This asymmetry is fundamental to all DNA replication

5. Leading Strand and Lagging Strand

Concept

Due to the antiparallel nature of DNA and the $5' \rightarrow 3'$ directionality of DNA polymerase:

- **Leading strand**: Synthesized continuously toward the replication fork
- **Lagging strand**: Synthesized discontinuously away from the replication fork

Why It Matters

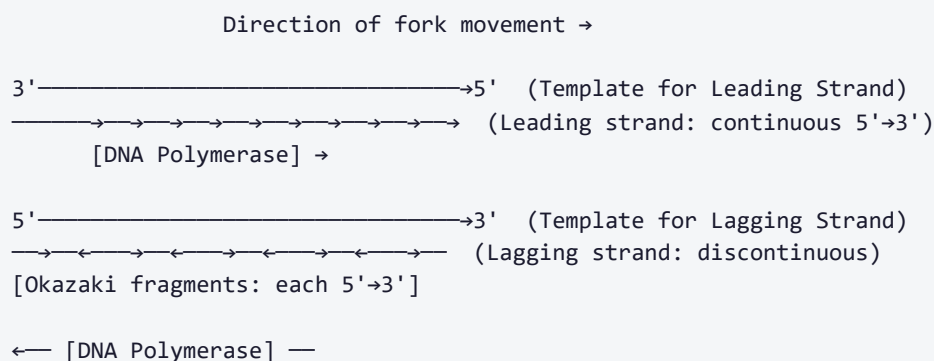
Understanding leading vs. lagging strand synthesis is essential for:

- Explaining Okazaki fragments
- Understanding why different polymerases may act on different strands
- Comprehending strand-specific mutation rates
- Understanding replication-associated diseases

Beginner Explanation

Because DNA polymerase can only work in one direction ($5' \rightarrow 3'$), one strand is easy to copy — just follow the unwinding fork. The other strand is harder — you have to copy it in short pieces as more template becomes available, like writing a sentence backwards by writing it in short segments.

Molecular-Level Explanation



Okazaki Fragments

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Size	1,000–2,000 nt	100–200 nt
RNA primer length	~10–12 nt	~8–12 nt
Processing	DNA Pol I removes primer, fills gap	RNase H + FEN1 remove primer; Pol δ fills gap
Ligation	DNA ligase (NAD ⁺)	DNA ligase I (ATP)

Why 5'→3' Direction?

The 5'→3' directionality of DNA synthesis has a deep chemical basis:

- DNA polymerase catalyzes a **nucleophilic attack** by the 3'-OH of the growing chain on the α-phosphate of the incoming dNTP
- This reaction releases pyrophosphate (PPi), which is subsequently hydrolyzed
- The energy from dNTP hydrolysis + PPi hydrolysis drives the reaction forward
- **There is no known mechanism** for 3'→5' polymerization in biology

Visual Explanation

LEADING STRAND SYNTHESIS

=====

Template: 3'-T-A-C-G-G-T-A-C-G-A-T-5'

New strand: 5'-A→T→G→C→C→A→T→G→C→T
 →→→→→ (continuous)

LAGGING STRAND SYNTHESIS (Okazaki fragments)

=====

Template: 5'-A-T-G-C-C-A-T-G-C-A-T-3'

Fragment 1: 5'-A-T-G-C-C→ (5'→3')

Fragment 2: 5'-A-T-G-C→ (5'→3')

Fragment 3: 5'-A-T-G→ (5'→3')

Each fragment is made 5'→3' but overall direction is ←

Common Misconception

Misconception: The lagging strand is synthesized $3' \rightarrow 5'$. **Reality:** Both strands are synthesized $5' \rightarrow 3'$. The lagging strand is synthesized as short fragments (Okazaki fragments) that are each made $5' \rightarrow 3'$, but the overall direction of synthesis on this strand is opposite to fork movement.

6. Directionality and Polarity

Key Terminology

Term	Meaning
5' end	End with a free phosphate group on the 5' carbon of the sugar
3' end	End with a free hydroxyl group on the 3' carbon of the sugar
Antiparallel	The two strands of DNA run in opposite directions
Template strand	The strand being read by DNA polymerase (read $3' \rightarrow 5'$)
Coding strand	The non-template strand (has same sequence as RNA product, $5' \rightarrow 3'$)
Processivity	Number of nucleotides added before polymerase dissociates
Processivity factor	Protein (e.g., sliding clamp) that increases processivity

The Central Rule

DNA is always synthesized in the $5' \rightarrow 3'$ direction. This is universal — no known exception exists in any organism.

Why This Matters

This single rule explains:

1. Why the lagging strand is discontinuous
2. Why Okazaki fragments exist
3. Why a primer is needed (provides the initial $3'$ -OH)
4. Why replication is asymmetric at the fork
5. Why different enzymes are needed for primer removal and gap filling

7. Overall Replication Workflow

Step-by-Step Mechanism

COMPLETE DNA REPLICATION WORKFLOW

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1. INITIATION
 - ↳ Recognition of origin of replication
 - ↳ Assembly of initiator complex
 - ↳ Local unwinding (melting) of double helix
2. PRIMING
 - ↳ Primase synthesizes short RNA primer
 - ↳ Provides free 3'-OH for DNA polymerase
3. ELONGATION
 - ↳ LEADING STRAND
 - ↳ DNA polymerase extends primer continuously
 - ↳ Follows helicase at the fork
 - ↳ Single primer needed
 - ↳ LAGGING STRAND
 - ↳ Multiple RNA primers synthesized
 - ↳ DNA polymerase synthesizes Okazaki fragments
 - ↳ Each fragment: ~100-200 nt (eukaryotes) or 1-2 kb (prokaryotes)
 - ↳ Direction: each 5'→3', overall away from fork
4. PRIMER REMOVAL AND GAP FILLING
 - ↳ Exonuclease removes RNA primers
 - ↳ DNA polymerase fills gaps with DNA
 - ↳ (Prokaryotes: Pol I; Eukaryotes: Pol δ + RNase H + FEN1)
5. LIGATION
 - ↳ DNA ligase seals nicks between Okazaki fragments
 - ↳ Creates continuous lagging strand
6. TERMINATION
 - ↳ Replication forks meet (circular chromosomes)
 - ↳ Termination sequences/structures
 - ↳ Daughter chromosomes separated
7. QUALITY CONTROL
 - ↳ Proofreading during synthesis (3'→5' exonuclease)
 - ↳ Mismatch repair after replication
 - ↳ Cell-cycle checkpoint verification

Quick Check

1. If you denature a completely replicated DNA molecule from the Meselson-Stahl experiment (generation 2, grown in ^{14}N after ^{15}N), what bands would you observe on a density gradient?
2. Why does DNA polymerase require a primer? What would happen if it didn't?
3. A student says: "The lagging strand is synthesized 3'→5'." Correct this statement.
4. List five requirements for DNA replication and explain the role of each.

5. If an organism's genome is 5×10^9 bp and replication proceeds at 50 nt/s per fork with 40,000 origins, approximately how long would S phase take? (Assume bidirectional replication.)

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Module 03: Prokaryotic DNA Replication

Learning Objectives

1. Describe the *E. coli* circular chromosome and *oriC*
2. Explain each step of prokaryotic replication initiation
3. Describe the roles of DnaA, DnaB, DnaC, and DnaG
4. Explain the mechanism of the Pol III holoenzyme
5. Describe leading and lagging strand synthesis in *E. coli*
6. Explain Okazaki fragment processing by Pol I
7. Describe termination using the Tus/Ter system
8. Understand replication fidelity mechanisms

Overview: *E. coli* as a Model System

Concept

Escherichia coli has been the primary model for understanding DNA replication since the 1960s. Its relatively small genome (4.6×10^6 bp), fast growth rate, and well-characterized genetics make it ideal for mechanistic studies.

Key Features

Feature	Value
Genome size	4,641,652 bp (circular)
Number of origins	1 (<i>oriC</i>)
Replication forks	2 (bidirectional)
Replication time	~40 minutes
Fork speed	~1,000 nt/second
Okazaki fragment size	1,000–2,000 nt
Number of genes	~4,300
Generation time (optimal)	~20 minutes

Why *E. coli*?

- Well-defined genetics (complete mutant collection)
- Reconstituted systems in vitro
- Many replication proteins identified and purified

- Structural information available for most components
- Clinical relevance (antibiotic targets)

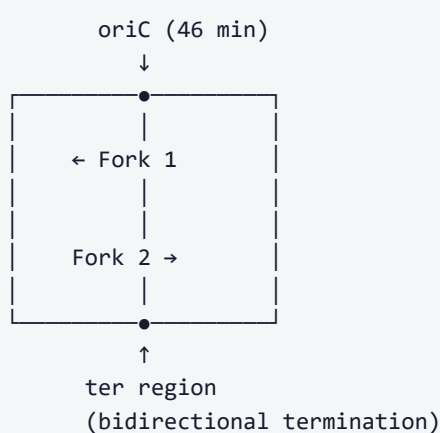
1. The *E. coli* Chromosome

Concept

The *E. coli* chromosome is a single circular double-stranded DNA molecule of approximately 4.6×10^6 base pairs. It is organized into a compact nucleoid structure.

Key Features

E. coli CHROMOSOME MAP (simplified)



Chromosome Organization

- **Circular:** No ends, no telomere problem
- **Supercoiled:** Negatively supercoiled (superhelical density ~ -0.06)
- **Nucleoid:** DNA organized with HU, H-NS, IHF, FIS proteins
- **Macrodomain structure:** Organized into topological domains

2. Origin of Replication: oriC

Concept

oriC is the single origin of replication in *E. coli*, located at 84.3 minutes on the genetic map.

Structure of oriC

oriC (245 bp)

Region 1 (13-mer repeats): AT-rich, easily melted

AT-rich	AT-rich	AT-rich
13-mer	13-mer	13-mer
(easy to unwind – fewer H-bonds)		

Region 2 (9-mer repeats / DnaA boxes):

R1	R2	R3	R4	R5
GATCTNTTNTNCAG (DnaA consensus)				
(DnaA binding sites)				

- R1, R2, R4, R5: High-affinity DnaA binding sites
- R3: Lower affinity (autoregulation)
- When enough DnaA-ATP accumulates → R3 binding → origin melts

Why oriC is Special

1. **AT-rich 13-mers:** A-T pairs have only 2 H-bonds (vs. 3 for G-C), so they melt more easily
2. **Multiple DnaA boxes:** Allow cooperative binding of initiator protein
3. **Small size:** 245 bp is sufficient for autonomous replication
4. **Unique in genome:** No other sequence works as efficiently as oriC

3. Initiation — Step by Step

Step 1: DnaA-ATP Accumulation

What happens:

- DnaA is the initiator protein
- DnaA binds ATP and ADP, but only DnaA-ATP is active for initiation
- The ratio of DnaA-ATP/DnaA-ADP increases during G1 phase
- When DnaA-ATP reaches a critical concentration, initiation begins

Which protein: DnaA (4×) **Why necessary:** Ensures replication initiates only once per cell cycle

Step 2: DnaA Binding to oriC

What happens:

- DnaA-ATP binds cooperatively to the 9-mer DnaA boxes (R1-R5)
- Forms a helical filament on DNA
- DnaA-ATP oligomerization induces positive supercoiling
- This destabilizes the adjacent AT-rich 13-mer region

Which protein: DnaA-ATP (20-30 molecules) **Why necessary:** Creates the initial opening (bubble) at the origin

Step 3: Origin Melting (Strand Separation)

What happens:

- The AT-rich 13-mer repeats melt (separate into single strands)
- Creates an open complex (replication bubble)
- 45 bp of DNA is initially unwound
- This is the rate-limiting step of initiation

Which protein: DnaA-ATP (indirect), thermal energy **Why necessary:** Provides single-stranded template for helicase loading

Step 4: DnaB (Helicase) Loading

What happens:

- DnaC (helicase loader) delivers DnaB to the melted origin
- DnaB is a hexameric ring helicase
- DnaC holds DnaB in an inactive state during loading
- DnaB encircles one strand of single-stranded DNA
- Once loaded, DnaC dissociates (ATP-dependent)

Which protein: DnaB (helicase, 2 hexamers loaded), DnaC (loader) **Why necessary:** DnaB unwinds DNA ahead of the replication fork

Step 5: Replication Bubble Formation

What happens:

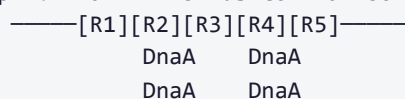
- Two DnaB helicases are loaded (one per fork)
- They move in opposite directions (bidirectional replication)
- DnaB uses ATP hydrolysis to unwind DNA (5'→3' direction on the strand it encircles)
- Creates two replication forks

Which protein: DnaB helicase (×2) **Why necessary:** Generates the Y-shaped replication fork structure

Visual Explanation: Initiation

INITIATION AT *oriC*

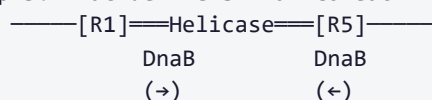
Step 1: DnaA-ATP binds to DnaA boxes



Step 2: AT-rich region melts



Step 3: DnaC delivers DnaB to each fork



Step 4: Bidirectional replication begins



4. Elongation — The Replisome

Concept

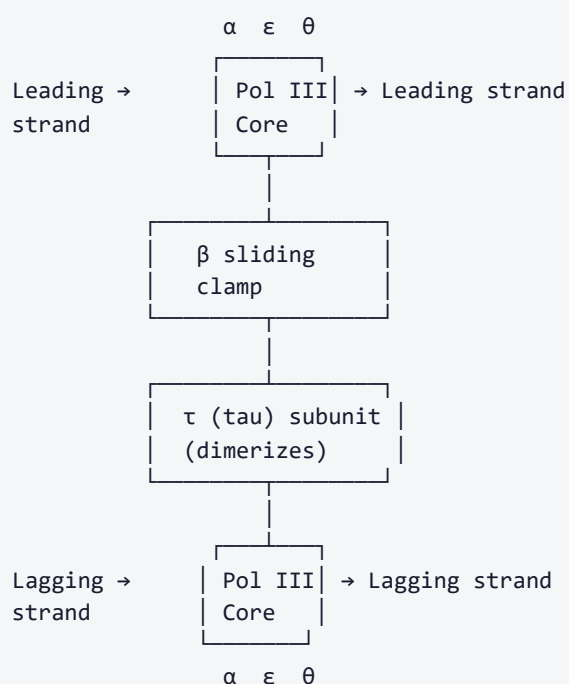
Once replication forks are established, the replisome — a complex molecular machine — carries out coordinated leading and lagging strand synthesis.

Components of the *E. coli* Replisome

Component	Function	Copies/Fork
DnaB helicase	Unwinds dsDNA (5'→3')	1 hexamer
DnaG primase	Synthesizes RNA primers	1–2
DNA Pol III	Synthesizes new DNA	2 (one per strand)
β sliding clamp	Increases processivity	2
Clamp loader (γ complex)	Loads β clamp	1 pentamer
SSB	Stabilizes ssDNA	Many
Topoisomerase II (Gyrase)	Relieves positive supercoiling	Ahead of fork

DNA Polymerase III Holoenzyme

The Pol III holoenzyme is the primary replicative polymerase in *E. coli*.

Pol III HOLOENZYME STRUCTURE

Subunits:

- α (dnaE): Polymerase activity (5'→3')
- ϵ (dnaQ): Proofreading exonuclease (3'→5')
- θ : Stimulates ϵ activity
- τ (dnaX): Dimerizes core, connects to helicase
- $\gamma/\delta/\delta'/\chi/\psi$: Clamp loader complex
- β (dnaN): Sliding clamp (processivity factor)

Pol III Core Function

Subunit	Gene	Function
α	dnaE	DNA polymerase (5'→3' synthesis)
ϵ	dnaQ	3'→5' exonuclease (proofreading)
θ	holE	Enhances ϵ stability

Combined error rate of Pol III core: $\sim 10^{-5}$ (before proofreading by ϵ : $\sim 10^{-7}$)

5. β Sliding Clamp**Concept**

The β sliding clamp is a ring-shaped protein that encircles DNA and tethers DNA polymerase to the template, dramatically increasing processivity.

Why It Matters

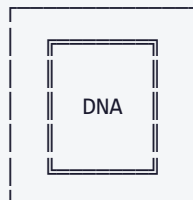
Without the β clamp:

- Pol III synthesizes only **10–20 nucleotides** before falling off

- With the β clamp: Pol III synthesizes **>50,000 nucleotides** without dissociating

Structure

β SLIDING CLAMP



← Ring-shaped homodimer
Inner diameter: $\sim 35 \text{ \AA}$
Fits dsDNA snugly

- Homodimer (β_2)
- 6-fold symmetry
- Slides freely along DNA (no sequence specificity)
- Interacts with Pol III core and clamp loader

Advanced Explanation

The β clamp is a **toroidal (ring-shaped)** protein with remarkable properties:

- It slides along DNA at the same speed as the polymerase
- It has **no intrinsic enzymatic activity** — it is purely a processivity factor
- Its structure is conserved: the eukaryotic equivalent (PCNA) has a similar ring shape but is a **trimer** (vs. dimer in prokaryotes)

6. Clamp Loader (γ Complex)

Concept

The clamp loader (also called the γ complex) uses ATP hydrolysis to open the β sliding clamp ring and load it onto primer-template junctions.

Components

Subunit	Function
γ (gamma)	ATPase, motor activity
δ (delta)	Opens β clamp
δ' (delta-prime)	Structural scaffold
χ (chi)	Interacts with SSB
ψ (psi)	Bridges γ and χ

Mechanism

1. γ complex binds ATP
2. γ -ATP complex binds β clamp → opens the ring

3. β -clamp-loaded complex binds primer-template DNA
4. ATP hydrolysis $\rightarrow$ β clamp closes around DNA
5. γ complex releases, leaving β clamp on DNA
6. Pol III core binds β clamp $\rightarrow$ polymerase is now processive

7. Leading Strand Synthesis

Step-by-Step Mechanism

LEADING STRAND SYNTHESIS

1. DnaB helicase unwinds DNA at the fork
2. SSB proteins coat the exposed single-stranded template
3. DnaG primase synthesizes a short RNA primer (~10-12 nt)
4. Clamp loader loads β clamp at the primer-template junction
5. Pol III core binds β clamp $\rightarrow$ now highly processive
6. Pol III extends primer continuously in 5' $\rightarrow$ 3' direction
7. Pol III follows DnaB helicase at the fork

Direction: same as fork movement $\rightarrow$

Speed: ~1,000 nt/second

Processivity: >50,000 nt (with β clamp)

Only ONE primer needed per fork

Key Point

The leading strand only needs **one primer** because synthesis is continuous. Once primed, Pol III follows the helicase around the entire chromosome.

8. Lagging Strand Synthesis — Okazaki Fragments

Concept

The lagging strand is synthesized discontinuously as short fragments called **Okazaki fragments** (named after Reiji and Tuneko Okazaki, who discovered them in the 1960s).

Why Okazaki Fragments Exist

The lagging strand template runs 5' $\rightarrow$ 3' in the direction of fork movement. Since DNA polymerase can only synthesize 5' $\rightarrow$ 3', synthesis on this strand must proceed **away from** the fork. As the fork unwinds more template, a new primer must be laid down and a new fragment synthesized.

Step-by-Step Mechanism

LAGGING STRAND SYNTHESIS (cycle repeats)

Fork movement →

3'=====5' (lagging template)

Step 1: Fork exposes new ssDNA

Step 2: DnaG primase synthesizes RNA primer (→ direction)

5'-rN-rN-rN-rN-rN-rN-rN-rN-rN-rN-3'

Step 3: Clamp loader loads β clamp

Step 4: Pol III synthesizes Okazaki fragment (5'→3', away from fork)

5'-rN-rN-dN-dN-dN-dN-dN-dN-dN-dN-3'

[RNA]-----[DNA]-----→

Step 5: Pol III reaches RNA primer of previous Okazaki fragment

Step 6: Pol III releases (collision with previous fragment)

Step 7: β clamp recycled to new primer

Step 8: Cycle repeats

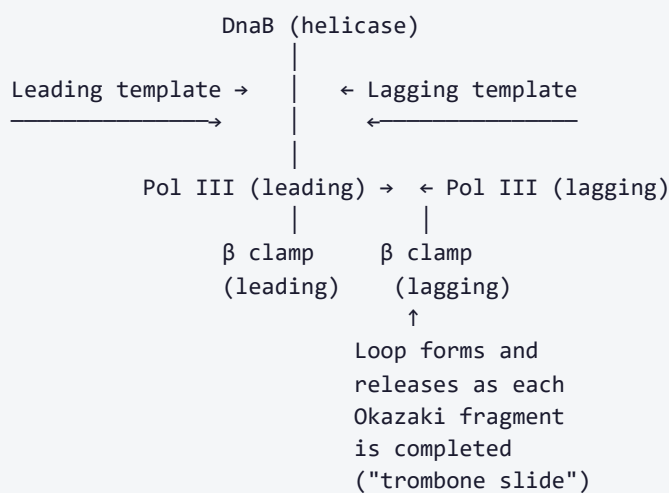
Okazaki fragment size: 1,000-2,000 nt

Each fragment: RNA primer (10-12 nt) + DNA (~1,000-2,000 nt)

The Trombone Model

The lagging strand template loops back so that both Pol III cores can move in the same physical direction at the fork:

THE TROMBONE MODEL



Common Misconception

Misconception: Okazaki fragments on the lagging strand are synthesized simultaneously. **Reality:** Only one Okazaki fragment is being actively synthesized at a time. The lagging strand Pol III cycles between primers, completing one fragment before starting the next.

9. Primer Removal and Gap Filling

Concept

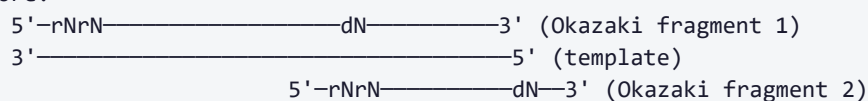
After Okazaki fragments are synthesized, the RNA primers must be removed and replaced with DNA to create a continuous strand.

Mechanism in *E. coli*

DNA Polymerase I (Pol I) performs primer removal and gap filling:

PRIMER REMOVAL BY POL I

Before:



Step 1: Pol I binds to 5' end of Okazaki fragment 2

(at the nick between fragments)

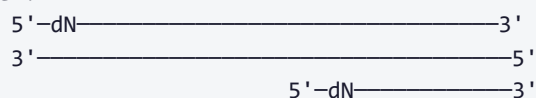
Step 2: Pol I uses 5'→3' exonuclease to remove RNA primer

Step 3: Pol I simultaneously fills gap with DNA (nick translation)

Step 4: RNA is degraded, replaced with DNA

Step 5: A nick remains between the two DNA fragments

After:



Nick at the junction → sealed by DNA ligase

Pol I — A Unique Polymerase

Activity	Direction	Function
5'→3' polymerase	5'→3'	Fills gaps (DNA synthesis)
3'→5' exonuclease	3'→5'	Proofreading
5'→3' exonuclease	5'→3'	Removes RNA primers (unique to Pol I among <i>E. coli</i> polymerases)

*The 5'→3' exonuclease activity of Pol I is critical — it is the only enzyme that can remove RNA primers ahead of the polymerase in *E. coli*.*

10. DNA Ligase

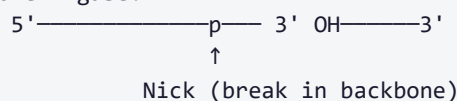
Concept

DNA ligase seals the nicks (breaks in the sugar-phosphate backbone) between Okazaki fragments after primer removal and gap filling, creating a continuous DNA strand.

Mechanism

DNA LIGASE REACTION

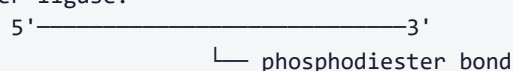
Before ligase:



Ligase mechanism:

1. Ligase binds NAD^+ (in *E. coli*) or ATP (in eukaryotes/viruses)
2. Ligase-AMP intermediate forms (AMP attached to lysine)
3. AMP transferred to 5'-phosphate at nick
4. 3'-OH attacks activated 5'-phosphate $\rightarrow$ phosphodiester bond
5. AMP released

After ligase:



Important Note

In *E. coli*, DNA ligase uses NAD^+ as a cofactor (not ATP). This is different from eukaryotic DNA ligases, which use ATP.

11. Topoisomerases and DNA Gyrase

Concept

As the helicase unwinds DNA ahead of the replication fork, positive supercoils accumulate. Topoisomerases relieve this torsional stress to prevent the DNA from tangling and breaking.

Why It Matters

Without topoisomerase activity:

- Positive supercoils accumulate ahead of the fork
- DNA becomes increasingly difficult to unwind
- Fork progression stops (fork stalling)
- DNA can break under torsional stress

Types of Topoisomerases in *E. coli*

Enzyme	Type	Mechanism	Function in Replication
DNA Gyrase	Type II	Cuts both strands, passes another duplex through, reseals	Introduces negative supercoils; relieves positive supercoils ahead of fork
Topoisomerase I	Type I	Cuts one strand, allows rotation, reseals	Relieves negative supercoils
Topoisomerase IV	Type II	Similar to gyrase	Decatenates (separates) daughter chromosomes after replication

DNA Gyrase — Unique to Bacteria

DNA gyrase is unique because it is the **only topoisomerase that can introduce negative supercoils** into DNA. It:

- Uses ATP hydrolysis
- Creates a double-strand break
- Passes another segment of DNA through the break
- Reseals the break
- Results in net introduction of negative supercoils

Clinical relevance: DNA gyrase is the target of **fluoroquinolone antibiotics** (ciprofloxacin, levofloxacin) — they trap the gyrase-DNA complex, causing lethal DNA breaks.

12. Complete Replication Fork — All Components

Visual Explanation

COMPLETE E. coli REPLICATION FORK

Direction of fork movement →

Topoisomerase/Gyrase
(relieves positive supercoiling)



← Parental dsDNA (ahead of fork)

↓ DnaB Helicase
(unwinds dsDNA)



LEADING STRAND (top):

5' → 3' template
← 3' → 5' new (continuous)
[Pol III + β clamp]

LAGGING STRAND (bottom):

3' → 5' template
← Fragment 1 → ← Fragment 2 → (discontinuous)
[RNA][DNA] [RNA][DNA]
[Pol III + β clamp]

AFTER the fork (lagging strand processing):

Pol I: removes RNA, fills gaps
DNA ligase: seals nicks
→ Continuous lagging strand

13. Bidirectional Replication

Concept

Replication from *oriC* proceeds bidirectionally — two replication forks move in opposite directions around the circular chromosome until they meet at the termination region.

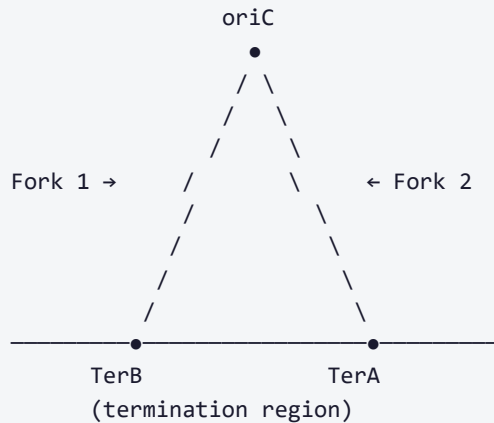
Why Bidirectional?

- Two DnaB helicases are loaded at *oriC* (one for each fork)
- Each fork has its own complete replisome

- Doubles the speed of genome replication (40 min instead of 80 min)

Replication Timing

E. coli CHROMOSOME REPLICATION



- Both forks travel at ~1,000 nt/second
- Complete replication: ~40 minutes
- Forks meet in the Ter region (opposite side from *oriC*)

14. Termination

Concept

Replication terminates when the two converging replication forks meet in the termination region, opposite to *oriC* on the circular chromosome.

The Tus/Ter System

E. coli uses a **polar replication fork trap** to ensure termination occurs in the correct region:

Tus/Ter SYSTEM

Ter sites: 10 sequences (TerA-TerJ) arranged as a "trap"

Tus protein: Binds Ter sequences

Tus binds Ter → creates a polar fork barrier:

TerB ← Fork 2 can pass through → allowed
Fork 1 ← blocked by Tus-TerB

TerA ← Fork 1 can pass through → allowed
Fork 2 ← blocked by Tus-TerA

Result: Both forks are trapped in the Ter region
→ They meet and terminate there

What Happens at Termination

1. **Forks converge:** The two replisomes meet

2. **Replication machinery disassembles:** Helicase, polymerase, etc. are removed
3. **Catenanes form:** The two daughter chromosomes are interlinked (like chain links)
4. **Topoisomerase IV decatenates:** Separates the interlinked circles
5. **Two daughter chromosomes:** Each is a complete circular molecule

Post-Termination Events

Event	Enzyme	Purpose
Fork convergence	—	Both replisomes meet
Replisome disassembly	Tus, proteases	Release replication machinery
Catenane formation	—	Topological consequence of bidirectional replication
Decatenation	Topoisomerase IV	Separates daughter chromosomes
Chromosome segregation	FtsK, MukBEF	Ensures each daughter cell gets one chromosome
Cell division	FtsZ ring	Cytokinesis

15. Replication Fidelity

Concept

DNA replication in *E. coli* achieves an extraordinary error rate of approximately **one mistake per 10^9 – 10^{10} nucleotides** incorporated. This is achieved through three layers of error correction.

Three Layers of Fidelity

REPLICATION FIDELITY

Layer 1: BASE SELECTION by Pol III

- Polymerase active site favors correct Watson-Crick pairs
- Geometric selection: only correct pairs fit
- Error rate: ~1 in 10^5 (10^{-5})

Layer 2: PROOFREADING by ϵ (3'→5' exonuclease)

- After incorrect nucleotide incorporation:
 - Polymerase stalls (mismatch distorts geometry)
 - DNA transfers from polymerase site to exonuclease site
 - Incorrect nucleotide excised (3'→5')
 - Correct nucleotide incorporated
- Error rate: improves to ~1 in 10^7 (10^{-7})

Layer 3: POST-REPLICATIVE MISMATCH REPAIR (MMR)

- MutS: recognizes mismatches and small insertion/deletion loops
- MutL: coordinates repair
- MutH: nicks the unmethylated (new) strand
- Exonuclease removes incorrect nucleotide from new strand
- Pol III resynthesizes
- Final error rate: ~1 in 10^9 – 10^{10} (10^{-9} to 10^{-10})

Strand Discrimination in MMR

How does MutH know which strand is new?

Methylation-based discrimination in *E. coli*:

- Dam methylase methylates adenine in GATC sequences
- Immediately after replication, the parental strand is methylated but the new strand is **not yet methylated** (hemimethylated state)
- MutH specifically nicks the **unmethylated** strand → ensures only the new strand is repaired

Note: In eukaryotes, strand discrimination in mismatch repair is thought to involve strand breaks (nicks) rather than methylation.

16. Replication Speed and Processivity

Speed

Parameter	Value
Fork speed	~1,000 nt/second
Replication time	~40 minutes
Total nucleotides per fork	$\sim 2.3 \times 10^6$
dNTP consumption	~2,000 nt/second (both forks combined)

Processivity

Without β clamp	With β clamp
10–20 nt	>50,000 nt
Pol III falls off easily	Pol III stays on for long stretches

dNTP Pool

- *E. coli* maintains a balanced pool of all four dNTPs
- Imbalanced dNTP pools increase mutation rates
- Ribonucleotide reductase (RNR) converts NDPs → dNDPs
- Nucleoside diphosphate kinase converts dNDPs → dNTPs

17. Special Topics in Prokaryotic Replication

Replication Coupled to Cell Division

In rapidly growing *E. coli* (doubling time < 40 min):

- A new round of replication initiates before the previous round finishes
- **Multiple replication forks** are present simultaneously

- **Nested replication forks:** Up to 4 forks can be active
- This allows generation times shorter than replication time

RAPID GROWTH – MULTIPLE ROUNDS

Generation time: 20 min

Replication time: 40 min

Solution: Initiate new rounds before previous rounds finish

Time 0: ← Fork 1 oriC Fork 2 →

Time 10: ← Fork 1 / \ Fork 2 →

 ← F3 F4 →

Time 20: Cell divides!

Each daughter has partially replicated chromosome

+ new initiation at oriC

Archaeal Replication (Brief Comparison)

Archaea share more similarities with eukaryotes than with bacteria:

Feature	Bacteria	Archaea	Eukaryotes
Origin	oriC	Multiple (similar to ORC)	Multiple ORIs
Initiator	DnaA	Orc1/Cdc6	ORC complex
Helicase	DnaB (3'→5')	MCM (5'→3')	MCM2-7 (5'→3')
Clamp	β (dimer)	PCNA (trimer)	PCNA (trimer)
Polymerase	Pol III	PolB/PolD	Pol α, δ, ε

Archaeal replication machinery is remarkably similar to eukaryotic, reflecting their shared evolutionary ancestry (both are in the supergroup Asgard/OPh group, or share common ancestry as prokaryotic/eukaryotic split).

Complete E. coli Replication Workflow

E. coli DNA REPLICATION – COMPLETE WORKFLOW

INITIATION

- |
- ↳ DnaA-ATP accumulates in the cell
- ↳ DnaA binds DnaA boxes in *oriC* (9-mer repeats)
- ↳ Cooperative binding + DNA wrapping
- ↳ AT-rich 13-mer region melts (open complex)
- ↳ DnaC loads DnaB helicase onto each strand
- ↳ DnaC dissociates (ATP-dependent)
- ↳ Two DnaB hexamers move in opposite directions
- |

PRIMING

- |
- ↳ DnaG primase interacts with DnaB
- ↳ DnaG synthesizes RNA primer (10-12 nt)
- ↳ Leading strand: one primer per fork
- ↳ Lagging strand: new primer every 1,000-2,000 nt
- |

ELONGATION

- |
- ↳ Clamp loader loads β sliding clamp at each primer
- ↳ Pol III core binds β clamp → processive complex
- ↳ LEADING STRAND:
 - | ↳ Pol III extends primer continuously (5'→3')
 - | ↳ Follows helicase
 - | ↳ No discontinuities
- |
- ↳ LAGGING STRAND:
 - | ↳ Pol III synthesizes Okazaki fragments (1-2 kb)
 - | ↳ Each fragment made 5'→3' (away from fork)
 - | ↳ Pol III cycles to new primer when it reaches previous fragment
 - | ↳ "Trombone model" – lagging template loops
- |
- ↳ PROOFREADING: ϵ subunit excises misincorporated bases
- |

PRIMER REMOVAL

- |
- ↳ Pol I uses 5'→3' exonuclease to remove RNA primers
- ↳ Pol I simultaneously fills gaps with DNA (nick translation)
- ↳ Gap is filled completely with DNA
- |

LIGATION

- |
- ↳ DNA ligase (NAD⁺) seals nicks between Okazaki fragments
- ↳ Creates continuous lagging strand
- |

TERMINATION

- |
- ↳ Forks converge in *Ter* region
- ↳ Tus-*Ter* system traps forks
- ↳ Replisomes disassemble
- ↳ Catenanes (interlinked daughter chromosomes) form
- ↳ Topoisomerase IV decatenates daughter chromosomes
- |

QUALITY CONTROL

- | → Proofreading (ϵ exonuclease) during synthesis
- | → Mismatch repair (MutS/L/H) after replication
- | → Strand discrimination via Dam methylation

COMPLETE: Two identical daughter chromosomes

Quick Check

1. What is the role of DnaA in initiation? What happens when DnaA cannot bind ATP?
2. Why is the 13-mer region of *oriC* AT-rich? What would happen if it were GC-rich?
3. A mutation in the ϵ subunit (*dnaQ*) eliminates proofreading. What would be the effect on mutation rate?
4. Explain why the β sliding clamp is essential for rapid replication but not for short DNA synthesis.
5. How does the Tus/Ter system ensure termination occurs in the correct region?
6. If DNA ligase is inactivated in *E. coli*, what would you observe?
7. Compare the roles of DnaB and DNA gyrase. Why are both necessary?

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Module 04: Eukaryotic DNA Replication

Learning Objectives

1. Describe the features of eukaryotic genomes relevant to replication
2. Explain the concept of replication licensing and how it prevents re-replication
3. Describe the roles of ORC, Cdc6, Cdt1, and MCM2-7
4. Explain the activation of the CMG helicase
5. Describe the functions of Pol α , Pol δ , and Pol ϵ
6. Explain the eukaryotic replisome and polymerase switching
7. Describe Okazaki fragment processing in eukaryotes
8. Explain chromatin remodeling during replication
9. Describe the end-replication problem and telomere biology
10. Explain S-phase regulation and replication checkpoints

Overview: Eukaryotic Replication Complexity

Key Differences from Prokaryotes

Feature	E. coli	Human
Genome size	4.6×10^6 bp	3.2×10^9 bp
Chromosome	1 (circular)	46 (linear)
Origins	1 (oriC)	30,000–50,000
Replication time	40 min	6–8 hours (S phase)
Fork speed	1,000 nt/s	~50 nt/s
Okazaki fragments	1,000–2,000 nt	100–200 nt
Chromatin	No (nucleoid)	Yes (nucleosomes)
Telomeres	Not needed	Required

Why Eukaryotic Replication is Different

1. **Genome size:** 3.2×10^9 bp would take ~53 hours to replicate from a single origin at 50 nt/s → multiple origins solve this
2. **Linear chromosomes:** Create end-replication problem → telomeres and telomerase
3. **Chromatin:** DNA is packaged into nucleosomes → must disassemble and reassemble during replication
4. **Cell-cycle regulation:** Must ensure each origin fires once per cell cycle → licensing system
5. **Multiple polymerases:** Different polymerases for different tasks → polymerase switching

1. Replication Licensing

Concept

Replication licensing is the mechanism that ensures each origin of replication fires **exactly once** per cell cycle. This prevents re-replication, which would cause gene amplification and genome instability.

Why It Matters

Re-replication can lead to:

- Gene amplification
- DNA damage (recombination between repeated sequences)
- Chromosome breaks
- Cell death or cancer

The Two-Step Model

REPLICATION LICENSING

G1 PHASE: LICENSING (assembly of pre-replication complex)

Origin → ORC binds → Cdc6 + Cdt1 recruit → MCM2-7 loaded
Origin is "licensed" (ready to fire)

- ✓ Multiple MCM2-7 complexes loaded per origin (bimolecular)
- ✓ Each origin loaded with 2 MCM helicases (one per fork)
- ✓ Origin is NOW competent to initiate replication

S PHASE: ACTIVATION (firing of licensed origins)

CDK + DDK phosphorylate MCM and other factors
→ Cdc45 and GINS loaded → CMG helicase forms
→ Origin fires → replication begins
→ ORC is displaced
→ Cdc6 is degraded
→ Cdt1 is degraded/inhibited

X Cannot re-license in S phase (CDK prevents new MCM loading)

G2/M PHASE: LICENSING CANNOT OCCUR

High CDK activity prevents new pre-RC assembly
→ No re-replication

Key Rule

CDK activity switches the system between licensing (low CDK → G1) and firing (high CDK → S/G2/M). This mutual exclusivity ensures no origin fires twice.

2. Origin Recognition Complex (ORC)

Concept

ORC is a six-subunit (Orc1-6) ATPase complex that serves as the landing pad for replication initiation factors at origins of replication.

Why It Matters

ORC is the **first protein** to bind at replication origins — it marks where replication will begin and recruits all subsequent factors.

Function

Property	Details
Subunits	Orc1-6 (AAA+ ATPase family)
ATP binding	Required for origin binding
ATP hydrolysis	Required for MCM loading
Conservation	Highly conserved from yeast to humans
Binding preference	More sequence-dependent in yeast (ARS consensus); less defined in metazoans

Mechanism

1. ORC binds origin DNA (ATP-dependent)
2. ORC recruits Cdc6 (ATPase, loads with ORC)
3. ORC-Cdc6 complex recruits Cdt1
4. Cdt1 delivers MCM2-7 double hexamer
5. ORC-Cdc6 hydrolyze ATP → MCM released from loader
6. Origin is licensed

Human vs. Yeast ORC

Feature	Yeast (<i>S. cerevisiae</i>)	Human
Origin definition	Strict ARS consensus sequence (ACS)	Loose; chromatin-dependent
ORC binding	Sequence-specific	Less sequence-specific
Origin efficiency	High, predictable	Variable, context-dependent
ORC subunits	Orc1-6	Orc1-6
Licensing in G1	Yes	Yes (with some differences)

3. Cdc6 and Cdt1

Cdc6

Cdc6 (Cell division cycle 6) is an AAA+ ATPase that works with ORC to load MCM2-7 helicases.

Function	Details
Binds ORC	Forms ORC-Cdc6 complex
ATP binding	Required for MCM loading
ATP hydrolysis	Releases MCM after loading
Regulation	Degraded in S phase by SCF ubiquitin ligase
Gene name	CDC6 (conserved)

Cdt1

Cdt1 (Chromatin licensing and DNA replication factor 1) is the direct MCM loader.

Function	Details
Binds MCM2-7	Directly interacts with MCM
Delivers MCM	Bridges ORC-Cdc6 to MCM2-7
Regulation	Inhibited by geminin in S/G2/M
Gene name	CDT1 (conserved)

Cdt1 Inhibition — The Geminin Pathway

GEMININ REGULATION OF LICENSING

G1 PHASE:

Cdt1 active → MCM loaded → Origin licensed

S PHASE:

Geminin accumulates (produced after licensing)

Geminin binds Cdt1 → inhibits MCM loading

Cdc6 degraded by SCF

→ No new licensing possible

M PHASE → G1:

Geminin degraded by APC/C (anaphase-promoting complex)

→ Cdt1 released → can license new origins in next G1

4. MCM2-7 Helicase

Concept

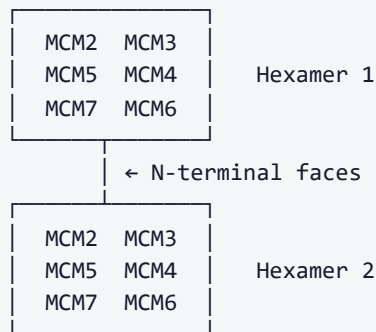
MCM2-7 (Minichromosome Maintenance 2-7) is the replicative helicase in eukaryotes. It is loaded as an inactive double hexamer at origins and activated during S phase to unwind DNA.

Why It Matters

MCM2-7 is the **engine** of the eukaryotic replisome — it provides the unwinding activity that drives fork progression. Understanding MCM is essential for understanding eukaryotic replication.

Structure

MCM2-7 DOUBLE HEXAMER (at licensed origin)



- Loaded as double hexamer (head-to-head)
- Each hexamer encircles one strand of dsDNA
- Inactive until activated by CDK/DDK + Cdc45/GINS
- When activated → CMG helicase forms

Properties

Property	Value
Subunits	MCM2, MCM3, MCM4, MCM5, MCM6, MCM7
ATPase	AAA+ family ATPase
Helicase direction	5'→3' (on the lagging strand template)
Processivity	Very high (can unwind > 10 kb)
Regulation	Inactive until phosphorylated by DDK + CDK

5. CMG Helicase — Activation

Concept

CMG (Cdc45-MCM-GINS) is the active replicative helicase in eukaryotes. It forms when Cdc45 and the GINS complex bind to the phosphorylated MCM2-7 hexamer.

Formation of CMG

CMG ACTIVATION (S-phase)

Step 1: DDK (Dbf4-dependent kinase) phosphorylates MCM2-7
 Step 2: CDK (Cyclin-dependent kinase) phosphorylates additional targets
 Step 3: Cdc45 binds (one per MCM hexamer)
 Step 4: GINS complex (Sld5-Psf1-Psf2-Psf3) binds
 Step 5: CMG helicase = Cdc45 + MCM2-7 + GINS
 Step 6: CMG is now active – begins unwinding

Before: Inactive MCM double hexamer

After: Two active CMG helicases (one per fork)

Properties of CMG

Property	Value
Components	Cdc45, MCM2-7, GINS
Helicase direction	5'→3' on the leading strand template
Unwinding mechanism	Encircles lagging strand template, translocates 5'→3'
Processivity	Very high
Regulation	Requires CDK + DDK for activation

GINS Complex

GINS stands for Go-Ichi-Ni-San (5-1-2-3 in Japanese), referring to the subunit numbers:

- Sld5 (GINS5)
- Psf1 (GINS1)
- Psf2 (GINS2)
- Psf3 (GINS3)

6. Replication Factors A (RPA)

Concept

RPA (Replication Protein A) is the eukaryotic single-strand binding protein. It binds and stabilizes single-stranded DNA at the replication fork.

Why It Matters

RPA:

- Prevents ssDNA from forming secondary structures (hairpins, G-quadruplexes)
- Protects ssDNA from nucleolytic degradation
- Coordinates multiple replication and repair factors
- Essential for fork stability

Properties

Property	Value
Subunits	RPA1 (70 kDa), RPA2 (32 kDa), RPA3 (14 kDa)
DNA binding	RPA1 has 4 OB-fold DNA binding domains
Binding mode	Binds ssDNA cooperatively
Equivalent in bacteria	SSB (single-strand binding protein)
Additional roles	DNA repair, checkpoint signaling

7. Eukaryotic DNA Polymerases

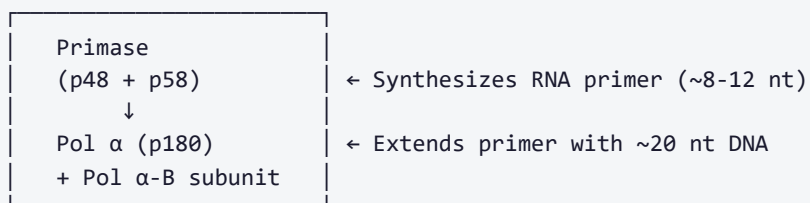
Overview

Eukaryotes use **three major replicative polymerases** and several specialized polymerases:

Polymerase	Function	Location	Size
Pol α/primase	Initiates synthesis; makes RNA-DNA primer	Nucleus	~350 kDa
Pol ϵ	Leading strand synthesis	Nucleus	~350 kDa
Pol δ	Lagging strand synthesis + repair	Nucleus	~240 kDa
Pol β	Base excision repair	Nucleus	39 kDa
Pol γ	Mitochondrial DNA replication	Mitochondria	~200 kDa
Pol η , κ , ι , Rev1	Translesion synthesis (error-prone)	Nucleus	Various
Pol ζ	Translesion synthesis (extension)	Nucleus	~350 kDa

Pol α /Primase Complex

POL α /PRIMASE COMPLEX



- Makes RNA-DNA hybrid primer (primosome)
- RNA primer: ~8-12 nt
- DNA extension: ~20 nt
- Total primer: ~20-30 nt
- Lacks 3'→5' exonuclease (NO proofreading)
- Initiates both leading and lagging strand synthesis
- Switched to Pol ϵ/δ after primer is made

Pol ϵ — Leading Strand

Property	Details
Function	Leading strand synthesis
Processivity	High (with PCNA)
Proofreading	Yes (3'→5' exonuclease)
Regulation	CDK phosphorylation activates
Proofreading mutant	Reduced fidelity → mutator phenotype

Pol δ — Lagging Strand

Property	Details
Function	Lagging strand synthesis + repair
Processivity	High (with PCNA)
Proofreading	Yes (3'→5' exonuclease)
Regulation	Pol δ is distributive without PCNA
Additional role	Okazaki fragment extension after Pol α

8. PCNA — Eukaryotic Sliding Clamp

Concept

PCNA (Proliferating Cell Nuclear Antigen) is the eukaryotic sliding clamp. It is a **homotrimer** forming a ring structure that encircles DNA and tethers polymerases to the template.

Comparison with Prokaryotic β Clamp

Feature	β Clamp (E. coli)	PCNA (Eukaryotes)
Structure	Homodimer	Homotrimer
Symmetry	6-fold	3-fold
Inner diameter	~35 Å	~35 Å
Outer diameter	~80 Å	~80 Å
Cofactor	—	— (loaded by RFC)
Recruitment of Pol	Via τ subunit	Direct interaction with Pol δ/ϵ

PCNA Interacting Proteins

PCNA interacts with many proteins via a **PCNA-interacting protein (PIP) motif**:

- DNA polymerases (Pol δ , Pol ϵ)
- FEN1 (Okazaki fragment processing)
- DNA ligase I

- CDK inhibitors (p21)
- Mismatch repair proteins
- Chromatin assembly factors (CAF-1)

9. RFC — Eukaryotic Clamp Loader

Concept

RFC (Replication Factor C) is the eukaryotic clamp loader that loads PCNA onto primer-template junctions.

Structure and Mechanism

Subunit	Function
RFC1 (p145)	Large subunit; ATPase
RFC2 (p40)	Small subunit
RFC3 (p37)	Small subunit
RFC4 (p36)	Small subunit
RFC5 (p33)	Small subunit

Mechanism:

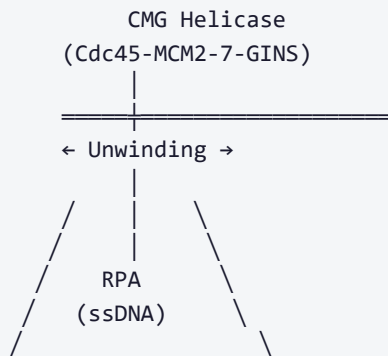
1. RFC binds ATP → opens PCNA ring
2. RFC-PCNA complex binds primer-template junction
3. ATP hydrolysis → PCNA closes around DNA
4. RFC releases
5. DNA polymerase binds PCNA

10. Eukaryotic Replication Fork

Complete Replisome

EUKARYOTIC REPLICATION FORK

Direction of fork movement →



LEADING STRAND:

5' → 3' (template)
 ← 3' → 5' (new DNA)
 Pol ϵ + PCNA

LAGGING STRAND:

3' → 5' (template)
 ← Fragment 1 → ← Fragment 2 →
 [RNA+DNA] [RNA+DNA]
 Pol α → Pol δ + PCNA

AFTER processing:

RNase H + FEN1 → remove RNA
 Pol δ → fill gaps
 DNA Ligase I → seal nicks
 CAF-1 → reassemble nucleosomes

11. Polymerase Switching

Concept

At each origin, synthesis is initiated by **Pol α /primase**, which makes an RNA-DNA primer. This is then replaced by a processive polymerase (**Pol ϵ** for leading strand, **Pol δ** for lagging strand). This transition is called **polymerase switching**.

Step-by-Step Mechanism

POLYMERASE SWITCHING

- Step 1: Pol α /primase synthesizes RNA-DNA primer
~8-12 nt RNA + ~20 nt DNA
- Step 2: RFC loads PCNA at the primer-template junction
(PCNA encircles the primer-template)
- Step 3: Pol ϵ (leading) or Pol δ (lagging) displaces Pol α
- High processivity polymerase binds PCNA
 - Pol α dissociates
- Step 4: Processive synthesis begins
- Leading: Pol ϵ synthesizes continuously
 - Lagging: Pol δ synthesizes Okazaki fragments
- Step 5: On lagging strand, when Pol δ reaches previous Okazaki fragment:
- Pol δ displaces the RNA primer (creates flap)
 - FEN1 cuts the flap
 - Pol δ continues

Why Polymerase Switching?

Pol α /primase	Pol ϵ/δ
Can start synthesis (has primase)	Cannot start synthesis
Low processivity	High processivity (with PCNA)
No proofreading	Has 3'→5' proofreading exonuclease
Error-prone	High fidelity

Analogy: Pol α is like a runner who starts a relay race (lays down the primer), then hands the baton to Pol ϵ/δ (the fast runner who completes the distance).

12. Okazaki Fragment Processing in Eukaryotes

Mechanism

EUKARYOTIC OKAZAKI FRAGMENT PROCESSING

After Pol δ synthesis:

```

5'—[RNA primer]—[DNA Okazaki fragment]—3' (fragment N+1)
3'—————5' (template)
5'—[RNA primer]—[DNA Okazaki fragment]—3' (fragment N)
  
```

Step 1: Pol δ encounters previous Okazaki fragment
→ Displaces some RNA as flap

Step 2: RNase H2 degrades most of the RNA primer
→ Leaves 1-2 ribonucleotides at junction

Step 3: FEN1 (Flap Endonuclease 1) removes:
- Remaining ribonucleotides
- Short DNA flaps displaced by Pol δ

Step 4: Pol δ fills the gap with DNA

Step 5: DNA Ligase I seals the nick

Result: Continuous DNA strand

FEN1 — A Special Endonuclease

Property	Details
Full name	Flap Endonuclease 1
Activity	5' flap endonuclease
Substrate	Displaced 5' flap structures
Mechanism	Cuts displaced flap, leaving 1 nt 5' overhang
Alternative	Dna2 helicase-nuclease can process longer flaps
Clinical	FEN1 mutations linked to cancer predisposition

13. Chromatin Remodeling During Replication

Concept

In eukaryotes, DNA is packaged into chromatin — a complex of DNA, histones, and other proteins. During replication, chromatin must be disassembled ahead of the fork and reassembled behind it.

Why It Matters

- Nucleosomes block replication machinery → must be disassembled
- Parental histones carry epigenetic marks → must be recycled

- New histones must be deposited to fill gaps
- Failure in chromatin reassembly → epigenetic instability

The Process

CHROMATIN DURING REPLICATION

Ahead of fork: Nucleosomes present (H3-H4 tetramers, H2A-H2B dimers)	At fork: Nucleosomes displaced, chaperoned by FACT	Behind fork: Nucleosomes reassembled with new + old histones
---	--	--

Key factors:

- FACT complex: Facilitates chromatin transcription (also facilitates replication through chromatin)
- CAF-1: Chromatin Assembly Factor 1 (deposits new H3-H4 tetramers)
- ASF1: Anti-silencing function 1 (histone chaperone for H3-H4)
- MCM2: Has histone chaperone activity (helps recycle parental histones)

Histone Recycling

Histone	Handling during replication
H3-H4 tetramer	Split between daughter strands; old + new mixed
H2A-H2B dimer	More dynamic; largely recycled
Epigenetic marks	Parental marks serve as template for new marks

Replication Timing

REPLICATION TIMING DURING S PHASE

Early S phase (first 2-3 hours):

- ↳ Euchromatin (active genes)
- ↳ High gene density regions
- ↳ GC-rich regions
- ↳ Highly accessible chromatin

Late S phase (last 2-3 hours):

- ↳ Heterochromatin (silent regions)
- ↳ Centromeric/peri-centromeric heterochromatin
- ↳ Telomeric heterochromatin
- ↳ Late-replicating gene-poor regions
- ↳ Facultative heterochromatin (X-inactivated chromosome)

S phase total: ~6-8 hours in human cells

Why Replication Timing Matters

1. **Gene expression:** Early replication correlates with active gene expression
2. **Epigenetic inheritance:** Late replication may facilitate epigenetic silencing
3. **Genome stability:** Late-replicating regions are more prone to rearrangements
4. **Development:** Replication timing programs change during differentiation

14. Licensing and Cell-Cycle Regulation

CDK Regulation of Licensing

CDK ACTIVITY AND LICENSING

G1 phase (low CDK activity):

- ✓ ORC can bind origins
- ✓ Cdc6 is stable
- ✓ Cdt1 is active
- ✓ MCM2-7 can be loaded
- Licensing occurs

S phase (high CDK activity):

- X CDK phosphorylates Cdc6 → degraded by SCF
- X CDK + geminin inhibit Cdt1
- X CDK phosphorylates ORC → reduced activity
- X CDK phosphorylates MCM → promotes activation
- No licensing, only firing

G2/M (high CDK):

- X Same restrictions as S phase
- No licensing

M → G1 transition:

- APC/C degrades geminin → Cdt1 released
- APC/C degrades CDK substrates
- Licensing can resume in next G1

Pre-Replication Complex (Pre-RC) Assembly

The pre-RC is the complete assembly at a licensed origin:

PRE-RC ASSEMBLY ORDER

1. ORC binds origin DNA (ATP-dependent)
2. Cdc6 binds ORC (ATP-dependent)
3. Cdt1 binds MCM2-7 and delivers to ORC-Cdc6
4. MCM2-7 loaded onto DNA (ATP hydrolysis)
5. Additional MCM2-7 loaded (cooperative)
6. Double hexamer forms (2 MCM2-7 per origin)
7. Cdc6 released, Cdt1 released/inhibited
8. Origin is LICENSED

Total MCM2-7 per origin: 4-5 double hexamers (in metazoans)
 = 8-10 MCM hexamers per origin
 = ensures robust initiation

15. Eukaryotic Termination

Concept

Unlike bacteria, eukaryotes do not have a defined termination sequence. Termination occurs when two converging replication forks meet.

Mechanism

EUKARYOTIC TERMINATION

- Step 1: Converging forks approach each other
 CMG helicases from opposite directions converge
- Step 2: CMG encounters incoming fork or DNA
 → CMG disassembly (requires p97/Cdc48 AAA+ ATPase)
 → CMG is ubiquitylated → extracted by p97
- Step 3: Remaining gaps filled by Pol δ
 → Okazaki fragment processing at junction
- Step 4: DNA Ligase I seals final nicks
- Step 5: No catenane problem in linear chromosomes
 (Telomeres prevent catenation)

CMG Disassembly — The End of the Road

Factor	Function
p97/Cdc48	AAA+ ATPase; extracts ubiquitylated CMG
Ubx5	p97 adaptor
SCF	Ubiquitylates MCM7
Sld2/Dpb11	Regulates termination

CMG disassembly is essential — persistent CMG would block subsequent processes (transcription, repair).

16. S-Phase Checkpoints

Concept

Cells monitor replication progression through checkpoint mechanisms that detect stalled forks and activate response pathways.

ATR-CHK1 Pathway

S-PHASE CHECKPOINT

Trigger: Stalled replication fork (long stretches of ssDNA)

Step 1: RPA-coated ssDNA accumulates at stalled fork
→ Excessive RPA-ssDNA is a danger signal

Step 2: ATRIP binds RPA-ssDNA → recruits ATR kinase
(ATR = Ataxia Telangiectasia and Rad3-related)

Step 3: ATR is activated by:
- TopBP1 (activator)
- Dna2 (processed DNA structures)

Step 4: ATR phosphorylates CHK1 kinase
→ CHK1 is the effector kinase

Step 5: CHK1 phosphorylates targets:
↳ Degrades Cdc25 phosphatase → inhibits CDK
↳ Stabilizes stalled forks
↳ Inhibits new origin firing
↳ Activates DNA repair pathways

Result:

- Replication slows down
- Damaged forks stabilized
- New origins suppressed (within same S phase)
- Time for repair before continuing

Fork Protection Complex

Component	Function
Timeless (Timestop)	Fork protection; recruits ATR
Claspin	Mediator; bridges MCM-ATRIP
Mrc1/Claspin	Stabilizes fork; promotes fork restart
14-3-3σ	Fork-associated protein

Complete Eukaryotic Replication Workflow

EUKEARYOTIC DNA REPLICATION – COMPLETE WORKFLOW

G1 PHASE: LICENSING

- |
- |→ Low CDK activity permits licensing
- |→ ORC binds replication origins
- |→ Cdc6 and Cdt1 recruited to ORC
- |→ MCM2-7 double hexamers loaded at origins
- |→ Origins are "licensed" (pre-RC assembled)
- |→ Multiple origins per chromosome (~30,000-50,000 total)
- |

S-PHASE ENTRY: ACTIVATION

- |
- |→ CDK activity rises (S-CDK: Cyclin E/A-CDK2)
- |→ DDK (Dbf4-dependent kinase) activated
- |→ DDK phosphorylates MCM2-7
- |→ S-CDK phosphorylates Sld2, Sld3
- |→ Cdc45 loaded → GINS loaded → CMG helicase forms
- |→ Origins FIRE → replication begins
- |→ Only licensed origins can fire (no re-replication)
- |

ELONGATION

- |
- |→ CMG helicase unwinds dsDNA
- |→ RPA binds and stabilizes ssDNA
- |→ Pol α /primase synthesizes RNA-DNA primer
- |→ RFC loads PCNA at primer-template junction
- |→ Polymerase switching: Pol α → Pol ϵ (leading) or Pol δ (lagging)
- |

|→ LEADING STRAND:

- |→ Pol ϵ + PCNA synthesizes continuously
- |→ Follows CMG helicase
- |→ Single initiation event per fork

|→ LAGGING STRAND:

- |→ New primer every 100-200 nt
- |→ Pol α /primase → Pol δ + PCNA
- |→ Okazaki fragments (~100-200 nt)
- |→ Pol δ extends primer, meets previous fragment

|→ CHROMATIN REMODELING:

- |→ FACT complex disassembles nucleosomes ahead of fork
- |→ MCM2 + Asf1 recycle parental H3-H4
- |→ CAF-1 deposits new H3-H4 tetramers
- |→ New nucleosomes assembled on daughter strands

OKAZAKI FRAGMENT PROCESSING

- |
- |→ RNase H2 removes most of RNA primer
- |→ FEN1 removes remaining RNA + short DNA flaps
- |→ Pol δ fills gaps with DNA
- |→ DNA Ligase I seals nicks
- |

FORK CONVERGENCE AND TERMINATION

- |
- |→ Two converging forks meet

```
|→ CMG helicases ubiquitylated
|→ p97/Cdc48 extracts CMG from DNA
|→ Remaining gaps filled by Pol δ
|→ DNA Ligase I seals final nicks
|
QUALITY CONTROL
|
|→ Proofreading by Pol ε and Pol δ (3'→5' exonuclease)
|→ Mismatch repair (MSH2/MSH6, MLH1/PMS2)
|→ ATR-Chk1 checkpoint monitors fork progression
|→ Fork protection complex stabilizes stalled forks
|
COMPLETE: Two identical daughter DNA molecules
           Each wrapped in properly assembled chromatin
```

Quick Check

1. Why do eukaryotes need multiple origins of replication while bacteria typically need only one?
2. What is the role of geminin in preventing re-replication?
3. Describe the difference between Pol α/primase and Pol ε in terms of function and proofreading ability.
4. Why is PCNA loaded at the primer-template junction and not at any position on ssDNA?
5. What would happen if Cdt1 cannot be inhibited by geminin?
6. Explain why the ATR-Chk1 checkpoint is important for genome stability.
7. Describe the chromatin reassembly process during eukaryotic replication.

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Module 05: Replication Fidelity and Repair

Learning Objectives

1. Explain the three layers of replication fidelity
2. Describe the mechanism of proofreading
3. Explain mismatch repair (MMR) in detail
4. Compare bacterial and eukaryotic mismatch repair
5. Understand the consequences of replication errors

1. Overview of Replication Fidelity

Concept

Replication fidelity refers to the accuracy with which DNA is copied. Multiple layers of error correction ensure that the final error rate is approximately **one mistake per 10^9 – 10^{10} nucleotides**.

The Error Cascade

Layer	Mechanism	Error Rate	Improvement
1. Base selection	Polymerase active site geometry	$\sim 10^{-5}$	—
2. Proofreading	3'→5' exonuclease	$\sim 10^{-7}$	100-fold
3. Mismatch repair	Post-replicative correction	$\sim 10^{-9}$ to 10^{-10}	100–1000-fold

Why Fidelity Matters

- Human genome: 3.2×10^9 bp
- Without any correction: $\sim 3.2 \times 10^4$ errors per cell division
- With all three layers: ~ 0.3 –3 errors per cell division
- Most remaining errors occur in non-coding DNA → few functional consequences

2. Base Selection — Layer 1

Concept

DNA polymerases discriminate between correct and incorrect nucleotides based on the geometry of Watson-Crick base pairs.

Mechanism

The polymerase active site acts as a **molecular caliper**:

- Correct base pairs (A-T, G-C) have a specific geometry
- Incorrect pairs (mismatches) have altered geometry
- Only correctly shaped pairs fit → catalysis proceeds
- Incorrectly shaped pairs → rejection (kinetic discrimination)

Two Mechanisms of Selection

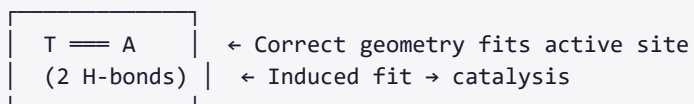
1. **Geometric selection:** Active site only accommodates correct pair geometry
2. **Induced-fit mechanism:** Correct binding triggers conformational change → catalysis proceeds;
incorrect binding → no conformational change → no catalysis

Visual Explanation

BASE SELECTION BY DNA POLYMERASE

Correct pair (A-T):

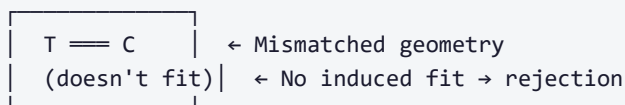
Template: T Incoming: A



→ dNTP incorporated ✓

Incorrect pair (A-C):

Template: T Incoming: C



→ dNTP rejected ✗

3. Proofreading — Layer 2

Concept

Proofreading is the immediate correction of misincorporated nucleotides by the 3'→5' exonuclease activity of DNA polymerase.

Mechanism

PROOFREADING BY 3'→5' EXONUCLEASE

Normal synthesis:

3'—Template—5'

5'—New DNA—→ 3' (growing end)

Misincorporation occurs:

3'—Template—5'

5'—New DNA—X→ 3' (X = wrong nucleotide)

↑

Mismatch detected!

Step 1: Polymerase stalls at mismatch

→ Distorted geometry at 3' end

Step 2: 3'→5' exonuclease domain activates

→ DNA partitions from polymerase site to exonuclease site

Step 3: Exonuclease removes the mismatched nucleotide

→ 3'→5' direction → removes last nucleotide(s)

Step 4: DNA returns to polymerase site

→ Correct nucleotide incorporated

3'—Template—5'

5'—New DNA—→ 3' (correct synthesis continues)

E. coli vs. Eukaryotic Proofreading

Feature	E. coli	Eukaryotic
Polymerase	Pol III (ε subunit)	Pol ε (leading), Pol δ (lagging)
Exonuclease domain	ε (dnaQ)	Built into Pol ε and Pol δ
Removal direction	3'→5'	3'→5'
Processing rate	~100 nt/s	~10 nt/s

Consequences of Proofreading Deficiency

Organism	Mutation	Phenotype
E. coli	dnaQ49 (ε mutant)	100-fold increase in mutation rate; often lethal
Mouse	Pol ε exonuclease dead	Embryonic lethal
Human	POLE/POLD1 exonuclease domain mutations	Cancer predisposition (Lynch-like syndrome)

4. Mismatch Repair (MMR) — Layer 3

Concept

Mismatch repair corrects errors that escape proofreading. It recognizes mismatched bases and small insertion/deletion loops that occur during replication.

Why It Matters

- Catches ~99% of errors that escape proofreading
- Reduces error rate from $\sim 10^{-7}$ to $\sim 10^{-9}$ – 10^{-10}
- **Defective MMR** → **Lynch syndrome (hereditary nonpolyposis colorectal cancer, HNPCC)**

E. coli Mismatch Repair — MutHLS System

E. coli MISMATCH REPAIR (MutHLS)

Step 1: RECOGNITION

- MutS homodimer (MutS_2) scans DNA
- Recognizes mismatches and small IDLs (insertion/deletion loops)
- MutS binds mismatch → ADP → ATP exchange → sliding clamp on DNA

Step 2: RECRUITMENT

- MutS recruits MutL (MutL homodimer)
- MutL acts as a molecular matchmaker
- MutL recruits MutH (endonuclease)

Step 3: STRAND DISCRIMINATION

- MutH nicks the UNMETHYLATED (new) strand
- E. coli Dam methylase methylates GATC sequences
- Immediately after replication: hemimethylated state
- Parental strand: methylated
- New strand: unmethylated (for ~2 minutes)
- MutH nicks unmethylated strand at nearest GATC

Step 4: EXCISION

- UvrD (helicase II) unwinds DNA from nick toward mismatch
- Exonuclease (ExoI, ExoVII, ExoX, or RecJ) degrades nicked strand
- Degrades past the mismatch

Step 5: RESYNTHESIS

- Pol III fills gap using methylated (parental) strand as template

Step 6: LIGATION

- DNA ligase seals the nick

Visual Explanation

MISMATCH REPAIR – STRAND DISCRIMINATION

Before repair:

```

5'-G-A-T-C-A-T-G-[X]-C-A-T-G-A-T-C-3' ← New strand (unmethylated)
3'-C-T-A-G-T-A-C-[Y]-G-T-A-C-T-A-G-5' ← Parent strand (methylated)
      ↑
    Mismatch (X-Y)
    e.g., A-C mismatch
  
```

Dam methylase methylates adenine in GATC:

- Parent strand: methylated (–N⁶methyl–A–)
- New strand: unmethylated (–A–)

MutH recognizes hemimethylated GATC:

→ Nicks unmethylated strand

After repair:

```

5'-G-A-T-C-A-T-G-[T]-C-A-T-G-A-T-C-3' ← Corrected new strand
3'-C-T-A-G-T-A-C-[A]-G-T-A-C-T-A-G-5' ← Parent strand
  
```

Eukaryotic Mismatch Repair

Eukaryotes lack MutH and do not use methylation for strand discrimination. Instead, strand breaks (nicks) mark the new strand.

EUKARYOTIC MISMATCH REPAIR

Step 1: RECOGNITION

- MSH2-MSH6 (MutS α): recognizes base-base mismatches and small IDLs (1 nt)
- MSH2-MSH3 (MutS β): recognizes larger IDLs (2-13 nt)

Step 2: RECRUITMENT

- MLH1-PMS2 (MutL α): endonuclease, nicks new strand
- MLH1-PMS1 (MutL β): additional function
- MLH1-MLH3 (MutL γ): trinucleotide repeat instability

Step 3: STRAND DISCRIMINATION

- No methylation system!
- Instead: nicks in the new strand serve as marks
- Okazaki fragment nicks (3' ends)
- PCNA-dependent strand discrimination
- RFC loads PCNA → MutL α recognizes PCNA orientation

Step 4: EXCISION

- EXO1 (5'→3' exonuclease) degrades nicked strand past mismatch
- RPA stabilizes remaining ssDNA

Step 5: RESYNTHESIS

- Pol δ fills gap (with PCNA)

Step 6: LIGATION

- DNA Ligase I seals nick

Comparison: Bacterial vs. Eukaryotic MMR

Feature	E. coli (MutHLS)	Eukaryotes (MutS/MutL)
MutS equivalent	MutS (homodimer)	MSH2-MSH6 (MutS α)
MutL equivalent	MutL (homodimer)	MLH1-PMS2 (MutL α)
Strand discrimination	Dam methylation (GATC)	Strand nicks (Okazaki fragments)
Helicase	UvrD	(role not fully defined)
Exonuclease	ExoI, VII, X, RecJ	EXO1
Polymerase	Pol III	Pol δ

Clinical Significance

Lynch Syndrome (HNPCC):

- Most common hereditary cancer syndrome
- Caused by germline mutations in MMR genes (MSH2, MLH1, MSH6, PMS2)
- ~1 in 300 individuals carry a mutation
- 50-80% lifetime risk of colorectal cancer
- Also increased risk of endometrial, ovarian, gastric cancers
- **Diagnosis:** Microsatellite instability (MSI) in tumors
- **Treatment:** May respond to immune checkpoint inhibitors (anti-PD-1)

5. Additional Repair Pathways

Base Excision Repair (BER)

Repairs small, non-helix-distorting lesions (oxidized bases, deaminated bases, abasic sites):

1. DNA glycosylase recognizes lesion → removes damaged base
2. AP endonuclease cuts backbone at abasic site
3. DNA polymerase (Pol β in eukaryotes) fills gap
4. DNA ligase seals nick

Nucleotide Excision Repair (NER)

Repairs bulky, helix-distorting lesions (UV-induced pyrimidine dimers, chemical adducts):

1. Damage recognition (XPC-RAD23B in eukaryotes)
2. Unwinding around damage (TFIIH helicases XPB, XPD)
3. Dual incision (XPF-ERCC1 5', XPG 3')
4. Gap filling (Pol δ/ϵ)
5. Ligation (ligase I)

Clinical: Defects in NER → Xeroderma Pigmentosum (extreme UV sensitivity, skin cancer)

6. Consequences of Replication Errors

Error Type	Frequency	Consequence
Point mutations (transitions)	$\sim 10^{-9}$ per bp per division	May change protein function
Point mutations (transversions)	$\sim 10^{-9}$ per bp per division	May change protein function
Frameshifts (IDLs)	Very rare with MMR	Protein truncation/function loss
Replication slippage	Common at repeats	Expansion/contraction of repeats

Mutation Rates

Organism	Mutation rate (per bp per generation)
E. coli	$\sim 5 \times 10^{-10}$
Yeast (<i>S. cerevisiae</i>)	$\sim 2 \times 10^{-10}$
Humans	$\sim 1 \times 10^{-8}$ (per cell division)
Per individual per generation	~ 70 new mutations

Quick Check

1. If you could only remove one layer of replication fidelity (base selection, proofreading, or mismatch repair), which would have the most severe consequences? Explain.
2. Why can't E. coli use methylation for strand discrimination if Dam methylase is inactivated?
3. A patient presents with microsatellite instability in their colorectal cancer. Which genes would you suspect are mutated? What syndrome does this suggest?
4. Explain how PCNA orientation helps eukaryotic cells distinguish the new strand from the parent strand during mismatch repair.
5. Why are point mutations at G:C $\rightarrow$ A:T transitions the most common type of spontaneous mutation?

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Module 06: Telomeres and Telomerase

Learning Objectives

1. Explain the end-replication problem
2. Describe telomere structure and function
3. Explain the mechanism of telomerase
4. Describe telomere regulation during aging and cancer
5. Understand shelterin complex function

1. The End-Replication Problem

Concept

Linear chromosomes cannot be fully replicated at their ends because:

1. RNA primers at the 5' end of the lagging strand are removed but cannot be replaced with DNA
2. There is no 3'-OH available for the final primer to be extended

This leads to progressive shortening of chromosomes with each cell division.

Why It Matters

- Without a solution → chromosomes shorten each division → eventual loss of essential genes → cell death
- Cancer cells must solve this problem to achieve unlimited proliferation
- Telomere shortening contributes to aging

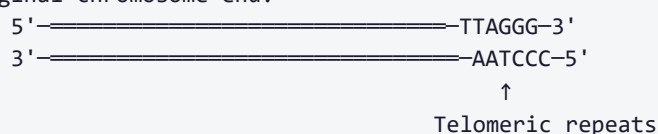
Beginner Explanation

Imagine copying a zipper from the middle. At each end, you can't copy the very tips because you need a starting point (primer). Each time you copy, a tiny bit is lost at the ends. Eventually, important parts of the zipper get lost.

Molecular-Level Explanation

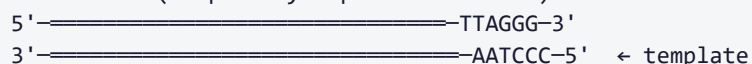
THE END-REPLICATION PROBLEM

Original chromosome end:

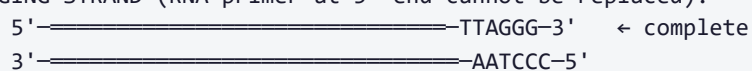


After one round of replication:

LEADING STRAND (completely replicated to end):



LAGGING STRAND (RNA primer at 5' end cannot be replaced):



Last Okazaki fragment is processed RNA primer at very 5' end of lagging strand is removed
→ GAP remains (cannot be filled)

Result: 3' overhang on parent strand
5' end of new strand is SHORTER

The Two Components of the Problem

Component	Cause	Effect
Primer removal	RNA primer at 5' end of lagging strand is removed → gap	5' end of new strand is shortened
No replacement	No upstream 3'-OH to extend from	Gap cannot be filled by DNA polymerase
Net result	Progressive shortening	~50–200 bp lost per division in humans

2. Telomere Structure

Concept

Telomeres are specialized nucleoprotein structures at the ends of linear chromosomes that solve the end-replication problem and protect chromosome ends from degradation and fusion.

Why It Matters

Telomeres serve two essential functions:

1. **Cap** chromosome ends to prevent DNA damage response activation
2. **Provide** a buffer of repetitive DNA that can be lost without losing essential genes

Human Telomere Sequence

HUMAN TELOMERE SEQUENCE

Sequence: TTAGGG (repeated thousands of times)

5'-TTAGGGTTAGGGTTAGGGTTAGGGTTAGGG...-3' (G-rich strand)

3'-AATCCCAATCCCAATCCCAATCCCAATCCC...-5' (C-rich strand)

Length: ~5-15 kb in humans at birth

Shortens with each division

Critical length: ~4-6 kb (triggers senescence)

Features:

- G-rich strand extends as 3' overhang (~150-200 nt)
- G-rich strand can form G-quadruplex structures
- C-rich strand can form i-motif structures
- T-loop structure: 3' overhang invades telomeric duplex
- D-loop: Displacement loop within T-loop

Telomere Structure — T-Loop

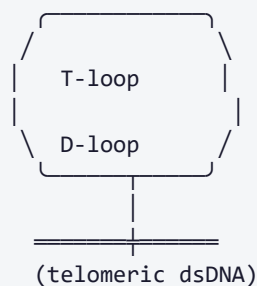
T-LOOP STRUCTURE

Linear view:



T-loop formation:

- 3' overhang folds back and **INVADES** the double-stranded telomeric DNA
- Creates a large loop (T-loop)
- Hides the chromosome end from being recognized as DNA damage



3. Shelterin Complex

Concept

Shelterin is a six-protein complex that binds telomeric DNA and regulates telomere length, structure, and protection.

Components

Protein	Function
TRF1	Binds double-stranded telomeric DNA; negative regulator of telomere length
TRF2	Binds ds telomeric DNA; essential for T-loop formation and end protection
POT1	Binds single-stranded G-rich overhang; protects 3' end
TIN2	Bridges TRF1, TRF2, and TPP1; central organizing factor
TPP1	Connects POT1 to TIN2; recruits telomerase
RAP1	Interacts with TRF2; transcriptional regulation

Shelterin Functions

1. **End protection:** Prevents chromosome ends from being recognized as DNA double-strand breaks
2. **T-loop formation:** TRF2 facilitates T-loop formation
3. **Telomere length regulation:** TRF1 provides negative feedback (more TRF1 → shorter telomeres → less telomerase access)
4. **Telomerase recruitment:** TPP1 recruits telomerase to telomeres
5. **ATR suppression:** Shelterin prevents ATR activation at telomeres (without shelterin → ATR recognizes telomeres as damage)

4. Telomerase

Concept

Telomerase is a ribonucleoprotein enzyme that extends telomeric DNA by adding TTAGGG repeats to the 3' end of chromosomes, counteracting the end-replication problem.

Discovery

- **Elizabeth Blackburn** and **Jack Szostak** (1982): Discovered that Tetrahymena telomeric sequences protect linear DNA in yeast
- **Carol Greider** (1984): Discovered telomerase activity in Tetrahymena
- Nobel Prize in Physiology or Medicine (2009): Blackburn, Greider, Szostak

Components of Human Telomerase

Component	Function
TERT (Telomerase Reverse Transcriptase)	Catalytic subunit; reverse transcriptase
TR/TERC (Telomerase RNA Component)	Contains template sequence (3'-CAAUCCCAAUC-5')
Dyskerin	Stabilizes TERC RNA
TCAB1	Cajal body localization
NAF1	Assembly factor

Telomerase Mechanism

TELOMERASE MECHANISM

Step 1: Telomerase binds to 3' overhang of telomere

POT1-TPP1 helps recruit telomerase

Step 2: Template alignment

TR template (3'-CAAUCCCAAUC-5') base-pairs with telomere 3' end

Telomere: 5'-TTAGGGTTAGGG-3'

|||||

TR template: 3'-CAAUCCC-5' (template region)

Step 3: Reverse transcription (extension)

TERT synthesizes DNA using TR as template

→ Adds TTAGGG repeats to 3' end

Step 4: Translocation

Telomerase moves to new 3' end

→ Repeats template alignment + extension

Step 5: Multiple rounds

→ Extends 3' overhang by many repeats

After telomerase:

Extended 3' overhang: 5'-TTAGGGTTAGGGTTAGGGTTAGGG...-3'

Step 6: Complementary strand synthesis

Primase/Pol α /primase synthesizes C-rich strand

→ Partial fill-in (does not fully replicate to end)

→ Maintains 3' overhang

Visual Explanation

TELOMERASE EXTENSION CYCLE

Before:

5'-TTAGGGTTAGGG-3' (G-rich strand)
3'-AATCCC—AATCCC-5' (C-rich strand, shorter)

Telomerase binds:

5'-TTAGGGTTAGGG-3'
3'-AATCCC—AATCCC-5'
 ||||||
TR template:-CAAUCCC-5'

Extension:

5'-TTAGGGTTAGGG-3'
3'-AATCCC—AATCCC-TTAGGG-5' (new TTAGGG added)

After multiple rounds:

5'-TTAGGGTTAGGG-3'
3'-AATCCC—AATCCC-TTAGGGTTAGGGTTAGGG-5'

C-rich strand filled in by conventional replication:

5'-TTAGGGTTAGGGTTAGGGTTAGGGTTAGGG-3'
3'-AATCCCAATCCCAATCCCAATCCCAATCCC-5'

5. Telomere Length Regulation

The Length-Setting Model

TELOMERE LENGTH HOMEOSTASIS

SHORT TELOMERES (< 5 kb):

- Few TRF1 molecules bound (less negative regulation)
- More telomerase access
- More extension
- Telomeres lengthen

LONG TELOMERES (> 10 kb):

- Many TRF1 molecules bound
- TIN2 recruits TRF1 → blocks telomerase
- Less telomerase access
- Telomere shortening continues (from end-replication problem)
- Equilibrium reached

BALANCE: Telomere length maintained at ~5-15 kb in humans

Telomerase Expression Pattern

Cell Type	Telomerase Activity	Telomere Status
Stem cells	Active (low level)	Maintained
Germ cells	Active	Maintained

Cell Type	Telomerase Activity	Telomere Status
Most somatic cells	Inactive	Shorten with age
Immune cells	Transiently activated	Partially maintained
Cancer cells (~85-90%)	Reactivated	Stabilized or lengthened
ALT cancer cells (~10-15%)	Inactive	Maintained by recombination

6. Telomeres and Aging

Concept

Progressive telomere shortening in somatic cells acts as a **biological clock** that limits cell division potential.

The Hayflick Limit

- **Leonard Hayflick (1961):** Normal human fibroblasts can divide ~40-60 times before entering senescence
- **Molecular basis:** Telomere shortening triggers senescence

Telomere Shortening and Senescence

TELOMERE SHORTENING → SENESCENCE

Birth: Telomeres ~10-15 kb
 ↓ (cell divisions)
 Age 30: Telomeres ~8-12 kb
 ↓
 Age 60: Telomeres ~6-10 kb
 ↓
 Critical length (~4-6 kb):
 → Shelterin can no longer form protective cap
 → Uncapped telomere recognized as DNA break
 → ATM/ATR checkpoint activation
 → p53/p21 pathway activation
 → Cell enters SENESCENCE (permanent growth arrest)
 → Or APOPTOSIS (cell death)

Aging Diseases

Disease	Mutation	Telomere Effect
Dyskeratosis congenita	DKC1, TERT, TERC	Short telomeres, premature aging
Hoyeraal-Hreidarsson syndrome	DKC1, TERT	Severe, early-onset
Revesz syndrome	TINF2	Cranial abnormalities, short telomeres

7. Telomeres and Cancer

Concept

Cancer cells must overcome the end-replication problem to achieve unlimited proliferation (immortality).

Telomerase Reactivation in Cancer

Mechanism	Frequency
TERT promoter mutations	~70% of melanomas, ~60% of glioblastomas
TERT amplification	Various cancers
TERT promoter methylation	Various cancers
TERT structural rearrangements	Neuroblastoma
ALT (Alternative Lengthening of Telomeres)	~10-15% of cancers

TERT Promoter Mutations

The most common telomerase-activating mutations in cancer:

- **C228T** and **C250T** mutations in the TERT promoter
- Create new transcription factor binding sites (ETS/GABP)
- Increase TERT transcription → more telomerase → telomere maintenance
- Found in: melanoma, glioblastoma, bladder cancer, hepatocellular carcinoma, thyroid cancer

Alternative Lengthening of Telomeres (ALT)

Some cancers maintain telomeres without telomerase:

- Uses **homologous recombination** between telomeric sequences
- Results in highly heterogeneous telomere lengths
- Characterized by **ALT-associated PML bodies (APBs)**
- Common in sarcomas, astrocytomas, some neuroendocrine tumors

Telomerase as Cancer Therapeutic Target

Strategy	Mechanism
Imetelstat (GRN163L)	Oligonucleotide complementary to TERC template; inhibits telomerase
BIBR1532	Small molecule inhibitor of TERT
G-quadruplex stabilizers	Stabilize G-quadruplex in telomeres → block telomerase access
Immunotherapy	Target TERT peptides presented on MHC

8. Clinical and Diagnostic Applications

Telomere Length Measurement

Method	Principle	Use
TRF Southern blot	Restriction digest + telomere probe	Gold standard; labor-intensive
qPCR	Telomere-specific primers	High-throughput; relative measurement
Flow-FISH	Flow cytometry + FISH	Single-cell resolution
TeSLA	Telomere shortening assay	Genome-wide telomere analysis

Diagnostic Applications

1. **Aging research:** Telomere length as biomarker
2. **Cancer prognosis:** TERT mutations as diagnostic markers
3. **Bone marrow failure:** Telomere length measurement for dyskeratosis congenita
4. **Pharmacogenomics:** TERT mutations predict response to immunotherapy

Quick Check

1. Explain why the lagging strand but not the leading strand is shortened at chromosome ends.
2. What would happen if shelterin were depleted from telomeres?
3. How does telomerase differ from conventional DNA polymerase?
4. A researcher discovers a tumor with highly heterogeneous telomere lengths and ALT-associated PML bodies. What telomere maintenance mechanism is this tumor using?
5. Why do most somatic cells not express telomerase? What is the evolutionary advantage?

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Module 07: Replication Stress and Checkpoints

Learning Objectives

1. Define replication stress and identify its causes
2. Explain replication fork stalling and recovery mechanisms
3. Describe the ATR-CHK1 checkpoint pathway
4. Explain fork reversal and restart
5. Describe the relationship between replication stress and genome instability

1. What is Replication Stress?

Concept

Replication stress (also called replication fork stress) is any condition that slows or blocks replication fork progression, leading to accumulation of stalled forks and abnormal DNA structures.

Why It Matters

Replication stress is a major cause of:

- **Genome instability** — the hallmark of cancer
- **Chromosome rearrangements** — translocations, deletions, amplifications
- **Common fragile sites** — regions prone to breakage during replication
- **Cancer development** — oncogene-induced replication stress drives tumorigenesis

Causes of Replication Stress

Cause	Mechanism	Example
DNA damage	Lesions block polymerase progression	UV photoproducts, oxidative lesions
Nucleotide depletion	Insufficient dNTPs	Hydroxyurea treatment, RNR deficiency
Replication-transcription conflicts	RNA polymerase blocks fork progression	Highly transcribed genes
R-loop accumulation	RNA:DNA hybrids stall forks	Transcription-replication conflicts
Secondary structures	G-quadruplexes, triplex DNA	GC-rich regions
Common fragile sites	Difficult-to-replicate regions	Large early-replicating genes
Topological stress	Excessive supercoiling	Topoisomerase inhibition
Oncogene activation	Aberrant origin firing, dNTP imbalance	Ras, Myc overexpression

Replication Fork Stalling

REPLICATION FORK STALLING

Normal fork progression:

- Fork moves smoothly at 50 nt/s

Stalled fork:

- Fork encounters obstacle (damage, structure, conflict)
- Fork stops
- ssDNA accumulates (exposed template)
- RPA coats ssDNA
- ATR checkpoint activated

If not resolved:

- Fork collapse
- Double-strand break
- Chromosome rearrangement
- Cell death or mutagenesis

2. Fork Reversal and Restart

Concept

When a replication fork stalls, it can undergo **fork reversal** (also called fork regression) — the newly synthesized strands anneal to each other, forming a "chicken foot" structure.

Fork Reversal Mechanism

FORK REVERSAL

Normal stalled fork:

```

3'=====→5' (lagging template)
←===== (new lagging strand)

5'=====→3' (leading template)
←===== (new leading strand, stopped at lesion)

X (blocked)
  
```

Fork reversal (chicken foot):

```

3'=====→5' (lagging template)
←===== (new lagging strand)
      \
       (new strands anneal)
      /
5'=====→3' (leading template)
←===== (new leading strand)
  
```

- Creates a structure resembling a 4-way junction
- Allows template switching or lesion bypass

Fork Restart Pathways

Pathway	Mechanism	When Used
Template switching	New strand uses intact sister chromatid as template	Lesion on template strand
Homologous recombination	RAD51/BRCA2-mediated strand invasion	DSB at collapsed fork
Repriming	Primase-Pol α re-primers downstream of lesion	Lesions that block polymerase
Translesion synthesis	Specialized polymerase bypasses lesion	Small lesions

3. The ATR-CHK1 Checkpoint

Concept

The ATR-CHK1 checkpoint is the primary S-phase checkpoint that responds to replication stress. It stabilizes stalled forks, inhibits new origin firing, and delays mitotic entry.

Pathway

ATR-CHK1 CHECKPOINT PATHWAY

1. STALLING EVENT

Fork stops → ssDNA exposed → RPA binds

2. SENSOR

ATRIP binds RPA-coated ssDNA

→ Recruits ATR kinase to stalled fork

3. ACTIVATION

TopBP1 (from 9-1-1 complex) activates ATR

→ ATR phosphorylates hundreds of substrates

4. SIGNALING

ATR phosphorylates CHK1

→ CHK1 is the effector kinase

5. EFFECTOR RESPONSES

↳ Stabilize stalled forks (prevent collapse)

↳ Suppress new origin firing (within same S phase)

↳ Delay S/M transition (prevent premature mitosis)

↳ Activate DNA repair pathways

↳ Slow cell-cycle progression

6. RESOLUTION

When stress resolved:

→ ATR/CHK1 deactivated

→ Forks restart or are repaired

→ Normal S-phase resumes

ATR vs. ATM

Feature	ATR	ATM
Primary trigger	ssDNA (replication stress)	DSBs

Feature	ATR	ATM
Checkpoint	S-phase	G1/S, G2/M
Key substrate	CHK1	CHK2
Partner	ATRIP	—
Yeast homolog	Mec1	Rad53/Ddc2

4. Common Fragile Sites

Concept

Common fragile sites (CFSs) are specific genomic regions that are particularly susceptible to forming gaps or breaks under replication stress. They are large, early-replicating regions that are difficult to replicate.

Why They Matter

- CFSs are breakpoints in cancer chromosomes
- Associated with large genes that take long to replicate
- Give insight into genome instability mechanisms
- May explain recurrent chromosomal rearrangements in cancer

Characteristics

Feature	Details
Number	~89 identified in human genome
Gene content	Often very large genes (> 1 Mb)
Replication timing	Late-replicating (despite early origins)
Expression	Many are transcribed → conflict with replication
Instability	Gaps/breaks under mild replication stress
Cancer relevance	Frequent breakpoints in tumors

5. Oncogene-Induced Replication Stress

Concept

Oncogene-induced replication stress (OIRS) is a major driver of genome instability in early cancer development. Oncogene activation creates replication stress through multiple mechanisms.

Mechanisms

ONCOGENE-INDUCED REPLICATION STRESS

Oncogene activation (e.g., Ras, Myc, Cyclin E):

1. ABERRANT ORIGIN FIRING
 - Excessive CDK activity
 - Fired origins that should be dormant
 - Fork collisions
 - Overwhelmed repair capacity
2. dNTP POOL IMBALANCE
 - Oncogenes upregulate RNR
 - Imbalanced dNTP pools
 - Increased misincorporation
 - Replication fork stalling
3. REPLICATION-TRANSCRIPTION CONFLICTS
 - Oncogene-driven transcription
 - Collisions between forks and RNA polymerases
 - R-loop formation
 - Fork breakage
4. INCOMPLETE LICENSING
 - Rapid S-phase entry
 - Not all origins licensed
 - Forks must travel further
 - More stalling events

Result: Genome instability → Cancer progression

Cancer Evolution

CANCER PROGRESSION AND REPLICATION STRESS

Normal cell → Oncogene activation → Replication stress

↓

Early cancer: Low-level stress, checkpoint response intact

→ Cell death or senescence (tumor suppression)

↓

Checkpoint adaptation/bypass:

→ Cells survive despite stress
 → Accumulate mutations
 → genome instability increases

↓

Advanced cancer:

→ High replication stress
 → Extensive genome instability
 → Chromosomal rearrangements
 → Drug resistance
 → Metastasis

6. Replication and DNA Damage Response Integration

Concept

Replication and DNA repair are tightly coupled. Damaged DNA is preferentially repaired during S phase when sister chromatids are available for homologous recombination.

Key Integration Points

Pathway	Replication Connection
Homologous recombination	Uses sister chromatid as template (only available after replication)
Translesion synthesis	Specialized polymerases bypass lesions at stalled forks
Template switching	Uses intact sister chromatid to bypass lesions
Fanconi anemia pathway	Repairs interstrand crosslinks during replication

Fanconi Anemia Pathway

- Repairs interstrand crosslinks (ICLs) that block both replication forks
- Requires coordination of nucleotide excision repair, translesion synthesis, and homologous recombination
- Mutations cause Fanconi anemia (bone marrow failure, cancer predisposition)
- Key proteins: FANCA, FANCB, BRCA2 (FANCD1), RAD51 (FANCR)

7. Replication Stress as Cancer Biomarker

Clinical Applications

Biomarker	Measurement	Clinical Use
γH2AX foci	Immunofluorescence	Marks DSBs from fork collapse
pRPA foci	Immunofluorescence	Marks ssDNA at stalled forks
53BP1 bodies	Immunofluorescence	Marks unrepaired DSBs
CIN (chromosomal instability)	Karyotyping, FISH	Genomic instability marker
CFSE	CFS expression	Fragile site expression

Therapeutic Implications

Strategy	Mechanism	Examples
Chemosensitization	Inhibit fork protection	ATR inhibitors (ceralasertib)
Synthetic lethality	Target checkpoint in deficient cells	ATR inhibitors in BRCA-mutant cancers
Replication stress induction	Increase stress beyond tolerance	Chk1 inhibitors (prexasertib)
PARP inhibitors	Block repair at stalled forks	Olaparib in BRCA-mutant cancers

Quick Check

1. List three causes of replication stress and explain how each causes fork stalling.
2. Describe the fork reversal mechanism and explain why it can be both protective and dangerous.
3. How does the ATR-CHK1 checkpoint prevent genome instability?
4. Explain how oncogene activation leads to replication stress.
5. Why are common fragile sites preferentially expressed in cancer cells?

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Module 08: Prokaryote vs. Eukaryote Comparison

Learning Objectives

1. Compare all major aspects of prokaryotic and eukaryotic replication
2. Identify conserved principles vs. species-specific differences
3. Understand why these differences exist
4. Apply comparative knowledge to predict replication features in unfamiliar organisms

Comprehensive Comparison Tables

1. Genome and Chromosome Structure

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Genome size	4.6 × 10 ⁶ bp	3.2 × 10 ⁹ bp
Chromosome number	1 (usually circular)	46 (linear)
Chromosome structure	Circular dsDNA	Linear dsDNA + telomeres
Supercoiling	Negative supercoiling	Nucleosome-based packaging
Organization	Nucleoid	Nucleus with chromatin
Associated proteins	HU, H-NS, IHF, FIS	Histones (H2A, H2B, H3, H4), H1
Packaging ratio	~1000:1	~10,000:1

2. Origins of Replication

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Number of origins	1 (oriC)	30,000–50,000
Origin size	245 bp	~100-400 bp (variable)
Sequence specificity	High (DnaA boxes, 13-mers)	Low (chromatin-dependent)
Initiator protein	DnaA	ORC (Orc1-6)
Origin regulation	DnaA-ATP/ADP ratio	CDK/DDK, geminin
Licensing system	None (single origin)	Pre-RC (ORC → Cdc6 → Cdt1 → MCM)
Firing control	Once per cell cycle	Once per cell cycle (licensing)

3. Replication Initiation

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Initiator	DnaA-ATP	ORC complex
Helicase loader	DnaC	Cdc6 + Cdt1
Replicative helicase	DnaB (3'→5')	CMG (Cdc45-MCM2-7-GINS) (5'→3')
Helicase orientation	Encircles lagging strand template	Encircles lagging strand template
Helicase loading	At origin (single complex)	At origin (double hexamer)
Activation	DnaA-dependent	CDK + DDK dependent
Bidirectional	Yes (2 forks)	Yes (2 forks per origin)

4. Elongation

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Replicative polymerase	Pol III holoenzyme	Pol ε (leading), Pol δ (lagging)
Primase	DnaG	Pol α/primase
Sliding clamp	β ₂ (homodimer)	PCNA (homotrimer)
Clamp loader	γ complex (5 subunits)	RFC (5 subunits)
SSB protein	SSB	RPA (RPA1-3)
Processivity factor	β clamp	PCNA
Fork speed	~1,000 nt/s	~50 nt/s
Replication time	~40 min	~6-8 hours (S phase)
Total nucleotides/fork	~2.3 × 10 ⁶	Variable (depends on inter-origin distance)

5. Okazaki Fragments

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Size	1,000–2,000 nt	100–200 nt
RNA primer length	~10–12 nt	~8–12 nt
Primer removal	Pol I (5'→3' exonuclease)	RNase H2 + FEN1
Gap filling	Pol I	Pol δ
Ligation	DNA ligase (NAD ⁺)	DNA ligase I (ATP)
Total fragments/fork	~2,300	~15,000

6. Proofreading and Repair

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Proofreading enzyme	ε subunit of Pol III	3'→5' exonuclease of Pol ε/δ

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Mismatch repair	MutS/MutL/MutH	MSH2-6/MLH1-PMS2
Strand discrimination	Dam methylation (GATC)	Strand nicks (PCNA-dependent)
BER	Multiple glycosylases	Pol β -dependent pathway
NER	UvrABC system	XPC/TFIIH/XPB/XPD
Final error rate	$\sim 10^{-9}$ – 10^{-10}	$\sim 10^{-9}$ – 10^{-10}

7. Topoisomerases

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Type II (ahead of fork)	DNA Gyrase	Topoisomerase II α
Type I	Topoisomerase I	Topoisomerase I
Decatenation	Topoisomerase IV	Topoisomerase II α
Unique feature	Gyrase introduces negative supercoils	No gyrase in eukaryotes
Drug target	Fluoroquinolones	Etoposide (Topo II), Camptothecin (Topo I)

8. Termination

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Termination site	Defined (Ter region)	Not defined (forks meet anywhere)
Termination system	Tus/Ter (polar trap)	None
Fork convergence	At Ter region	Wherever forks meet
Post-termination	Decatenation (Topo IV)	CMG disassembly (p97/Cdc48)
Catenane problem	Yes (circular chromosomes)	Minimal (linear chromosomes)
Chromosome segregation	FtsK, MukBEF	Cohesin, condensin, Smc

9. Telomeres

Feature	Prokaryotes	Eukaryotes
Telomere presence	No (circular chromosome)	Yes (linear chromosomes)
Telomere sequence	N/A	TTAGGG (humans)
Telomerase	No	Yes (TERT + TERC)
End-protection	N/A	Shelterin complex
End-replication problem	No (circular)	Yes (linear)

10. Cell-Cycle Regulation

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Cell cycle	Overlapping (no discrete phases)	Discrete (G1, S, G2, M)
Division control	FtsZ ring, Min system	Cyclin-CDK oscillations
Checkpoint	SOS response (minor)	ATR-Chk1, ATM-Chk2 (extensive)
Re-replication prevention	Overlapping initiation (no licensing)	Licensing system (CDK/Geminin)
Growth rate control	Growth rate → initiation frequency	Developmental → S-phase timing

11. Chromatin/Chromosome Dynamics

Feature	Prokaryotes (E. coli)	Eukaryotes (Human)
Chromatin structure	Nucleoid (HU, H-NS)	Nucleosomes + higher-order
Nucleosome disassembly	N/A	FACT complex, histone chaperones
Nucleosome reassembly	N/A	CAF-1, ASF1
Histone recycling	N/A	MCM2, FACT
Epigenetic inheritance	Limited (DNA methylation)	Extensive (histone marks, DNA methylation)
Replication timing	Not applicable	Early vs. late S phase

12. Speed and Efficiency

Parameter	E. coli	Human
Fork speed	1,000 nt/s	50 nt/s
Genome size	4.6×10^6 bp	3.2×10^9 bp
Origins	1	~30,000
Replication time	40 min	6-8 hours
Okazaki fragments	1-2 kb	100-200 bp
Polymerases involved	Pol I, Pol III	Pol α , Pol δ , Pol ϵ
Sliding clamp	β_2 (dimer)	PCNA (trimer)

Conservation of Replication Mechanisms

What is Conserved?

Despite billions of years of evolution, certain features are remarkably conserved:

1. **Semiconservative replication** — universal
2. **5'→3' synthesis** — universal
3. **Need for primers** — universal
4. **Sliding clamp** — ring-shaped processivity factor (prokaryotic β_2 → eukaryotic PCNA)

5. **Clamp loader** — AAA+ ATPase machine
6. **3'→5' proofreading exonuclease** — conserved in replicative polymerases
7. **Okazaki fragments** — discontinuous lagging strand synthesis
8. **SSB/RPA** — single-strand binding proteins
9. **Helicase** — ring-shaped hexameric motor
10. **Topoisomerase** — relieve torsional stress

What Differs?

Feature	Reason for Difference
Multiple origins	Larger genome needs faster replication
Licensing system	Prevents re-replication of large genome
Chromatin	Packaging of large genome in nucleus
Telomeres	Linear chromosomes need end protection
Slower fork speed	Nucleosome obstacles, more complex regulation
More polymerases	Specialized functions, chromatin context
Complex checkpoints	Larger genome, more at stake

Archaeal Replication — An Evolutionary Bridge

Archaea share features with both bacteria and eukaryotes:

Feature	Bacteria	Archaea	Eukaryotes
Genome	Circular	Circular (usually)	Linear
Origins	oriC (single)	Multiple (ORC-like)	Multiple (ORC)
Initiator	DnaA	Orc1/Cdc6	ORC complex
Helicase	DnaB (3'→5')	MCM (5'→3')	CMG (5'→3')
Clamp	β_2 (dimer)	PCNA (trimer)	PCNA (trimer)
Clamp loader	γ complex	RFC	RFC
Polymerase	Pol III	PolB/PolD	Pol $\alpha/\delta/\epsilon$
Primase	DnaG	PriS/PriL (eukaryotic-type)	Pol α /primase
MMR	MutS/MutL/MutH	MSH homologs	MSH2-6/MLH1-PMS2

*Archaeal replication machinery is **eukaryotic-like**, reflecting their shared evolutionary ancestry. This is one of the strongest arguments for the three-domain classification of life.*

Quick Check

1. Why do eukaryotes need multiple origins while bacteria need only one?
2. A researcher discovers a new organism with a circular genome of 2×10^7 bp. Would you predict it has one or multiple origins? Explain your reasoning.
3. Compare and contrast the sliding clamps of *E. coli* and humans. What aspects are conserved?
4. Why is the Tus/Ter system unnecessary for eukaryotic chromosomes?
5. How does the presence of chromatin in eukaryotes complicate replication?

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Module 09: Experimental Methods

Learning Objectives

1. Describe major experimental approaches to study DNA replication
2. Understand in vitro reconstitution of replication
3. Explain how replication origins are mapped
4. Describe fiber labeling and DNA combing
5. Understand the uses and limitations of each technique

1. In Vitro Replication Systems

Concept

In vitro reconstitution involves purifying all necessary replication proteins and combining them with DNA template to study replication mechanisms in a test tube.

E. coli In Vitro System

The complete E. coli replication machinery has been reconstituted from purified components:

Component	Source
DnaB helicase	Overexpressed, purified
DnaG primase	Overexpressed, purified
Pol III holoenzyme	Overexpressed, purified
SSB protein	Overexpressed, purified
β sliding clamp	Overexpressed, purified
γ complex	Overexpressed, purified
Gyrase	Overexpressed, purified
oriC plasmid	Cloned

Experiments Possible

1. **Origination:** Show that replication initiates at oriC
2. **Bidirectionality:** Demonstrate two forks from a single origin
3. **Processivity:** Measure polymerase processivity with/without β clamp
4. **Fidelity:** Measure error rates under different conditions
5. **Kinetics:** Measure fork speed, dNTP incorporation rates

2. 2D Gel Electrophoresis (BrdU/BrdC Labeling)

Principle

Bromodeoxyuridine (BrdU) is a thymidine analog that is incorporated into DNA during replication. It allows detection and analysis of replicating DNA.

Applications

- **Pulse labeling:** Short BrdU pulses label replicating DNA
- **Density gradient separation:** BrdU-containing DNA is heavier
- **Immunodetection:** Anti-BrdU antibodies detect labeled DNA

3. DNA Fiber Analysis

Concept

DNA fiber analysis is a powerful technique for measuring replication fork dynamics at single-molecule resolution.

DNA Combing

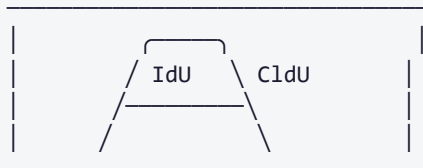
DNA COMBING TECHNIQUE

- Step 1: Label cells with nucleotide analogs
- IdU (iododeoxyuridine) – pulse 1
 - CldU (chlorodeoxyuridine) – pulse 2
- Step 2: Extract genomic DNA
- Gently extract high-molecular-weight DNA
- Step 3: Mount on glass coverslips
- DNA molecules attach to silanized glass
- Step 4: Stretch DNA by combing
- Pulling force straightens DNA molecules
 - ~2-3 kb/ μ m stretching ratio
- Step 5: Immunostaining
- Anti-BrdU antibody (different clones for IdU vs CldU)
 - Red signal = IdU (pulse 1)
 - Green signal = CldU (pulse 2)
- Step 6: Analyze fiber patterns
- Measure track lengths = fork speed
 - Count tracks = origin usage
 - Analyze fork asymmetry

DNA Fiber Interpretation

FIBER PATTERNS

REPLICATION BUBBLE:



→ Origin fired between pulses

TWO FORKS (bidirectional):

CldU ← IdU → → IdU ← CldU

→ Two forks moving in opposite directions

FORK COLLISION:

CldU ← IdU → IdU ← CldU

→ Forks converged

DORMANT ORIGIN:

CldU ← IdU → ← IdU CldU

→ One fork fired, other dormant

4. Okazaki Fragment Analysis

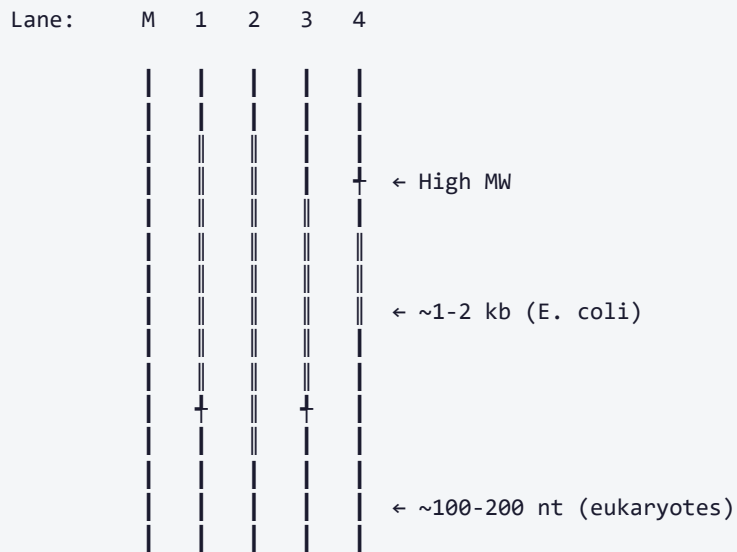
Method

Analyze Okazaki fragments by:

1. Labeling nascent DNA with [α - 32 P]dNTPs
2. Denaturing alkaline gel electrophoresis
3. Visualizing short nascent DNA fragments

Expected Results

ALKALINE GEL – OKAZAKI FRAGMENTS



M = size markers
 1 = Wild type (normal fragment size)
 2 = Ligase mutant (fragments persist)
 3 = Pol I mutant (fragments accumulate)
 4 = Small fragments (eukaryote sample)

5. Replication Origin Mapping

Methods for Origin Identification

Method	Resolution	Genome-wide?	Principle
ARS analysis (yeast)	Sequence-level	No	Clone fragments, test autonomous replication
Replication bubble mapping	~1-2 kb	Yes	2D gel electrophoresis
Origin mapping by microarray	~1 kb	Yes	Nascent strand abundance
Repli-seq	~100 bp	Yes	FACS + sequencing of early/late replicating DNA
Okazaki fragment sequencing	~100 bp	Yes	Sequence Okazaki fragments to map origins
STN-seq	~50 bp	Yes	Strand-specific Okazaki fragment sequencing

Repli-seq (Replication Timing Sequencing)

REPLI-SEQ WORKFLOW

1. Pulse-label cells with BrdU
2. Synchronize or sort cells by DNA content
3. FACS-sort early-S vs. late-S cells
4. Isolate BrdU-labeled (replicating) DNA
5. Sequence
6. Map peaks → early origins
7. Map valleys → late origins

Result: Genome-wide replication timing profile

6. Chromatin Immunoprecipitation (ChIP) for Replication

Applications

Target	ChIP Application
ORC subunits	Map origin binding sites
MCM2-7	Map licensed origins
PCNA	Map active replication forks
Pol ϵ / Pol δ	Map active polymerases
Cdc45	Map active CMG helicases
Histone modifications	Map replication timing domains

7. Fluorescence Microscopy

Applications

Technique	Use
GFP-tagged replication proteins	Visualize replication factories in live cells
Replication foci	Count and locate active replication sites
FRAP	Measure protein dynamics at replication forks
FISH	Map specific sequences relative to replication timing
STORM/PALM	Super-resolution imaging of replication factories

Replication Foci

REPLICATION FOCI IN S-PHASE NUCLEI

Early S phase:



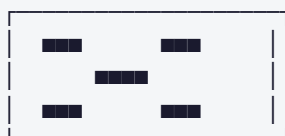
Small, numerous foci
(euchromatin replication)

Mid S phase:



Intermediate foci
(transitioning)

Late S phase:



Few large foci
(heterochromatin, centromeres)

8. CRISPR/Cas9 Approaches

Applications

Approach	Use
Origin deletion	Remove specific origins → study consequences
Origin replacement	Swap origins between locations
Protein tagging	Endogenous tagging of replication proteins
Gene knockout	Knock out replication genes (essential genes: use depletion)
Base editing	Introduce specific mutations in replication genes

Quick Check

1. What are the advantages and limitations of in vitro reconstitution vs. in vivo studies of replication?
2. How would you use DNA fiber analysis to determine if a drug slows replication fork speed?
3. Design an experiment to determine whether a specific protein is required for replication initiation vs. elongation.
4. Explain how Repli-seq can be used to map genome-wide replication timing.
5. How does ChIP-seq for MCM2-7 differ from ChIP-seq for PCNA in terms of what they tell you about replication?

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Module 10: Data Analysis and Interpretation

Learning Objectives

1. Interpret replication-related molecular biology data
2. Perform basic bioinformatics analysis of replication sequences
3. Analyze replication timing data
4. Interpret DNA fiber analysis results
5. Connect computational and experimental approaches

1. Analyzing Replication Origins Bioinformatically

Exercise 1: Identifying DnaA Boxes in *E. coli* oriC

Input:

E. coli oriC sequence (245 bp):

```
ATCTGGTAAAATTATTTACAGCGAAATTTATCAACTATCGTTAACTATCGTTAACGATTGAA
ATGATCGTCAAAACGATCGTTACATCAACTATCGTTAATTGATCGTTAACTATCGTTAACGATC
CGATCAACGATCATCGAACGATCAACGATCAATTCGATCATCAACGATCATCAACGATCGTAA
CAATCGTCGAATCAACGATCGTTCAGTTAACTATCGTTAACTATCGTTAACGATTCGATCATC
```

Task: Search for DnaA box sequences (GATCTNTTNTNCAG)

Expected Workflow:

1. Identify the consensus sequence for DnaA boxes
2. Scan the sequence for matches (allowing 1-2 mismatches)
3. Note positions of R1-R5 boxes
4. Identify the 13-mer AT-rich region

Key Results:

- R1-R5 DnaA boxes found at expected positions
- 13-mer AT-rich region identified upstream of DnaA boxes
- The arrangement allows cooperative DnaA binding

Exercise 2: Predicting Replication Origins in a Novel Bacterial Genome

Task: Given a 5 Mb bacterial genome sequence, predict likely oriC location

Approach:

1. Look for DnaA box consensus sequences
2. Identify AT-rich regions near DnaA boxes
3. Map基因 density (genes point away from oriC)

4. Use GC-skew analysis (G/C ratio changes at origin and terminus)

GC-Skew Analysis:

$$\text{GC-skew} = (G - C) / (G + C)$$

For a circular chromosome:

- GC-skew changes sign at origin and terminus
- Plot GC-skew along genome → peaks/valleys indicate oriC/ter

2. Analyzing Okazaki Fragment Data

Exercise: Interpreting Okazaki Fragment Size Data

Simulated Data:

Experiment: Alkaline gel analysis of Okazaki fragments

Lane 1: Wild type *E. coli*
→ Band centered at ~1,500 nt

Lane 2: *dnaG* mutant (primase mutant)
→ Band at ~3,000 nt (larger fragments)

Lane 3: ligase mutant (*E. coli* ligase⁻)
→ Smear of fragments (not joined)

Lane 4: Human cell extract
→ Band centered at ~150 nt

Lane 5: Human cells + FEN1 inhibitor
→ Slightly larger fragments + accumulation of flaps

Questions:

1. Why are Okazaki fragments larger in the *dnaG* mutant?
2. Why does the ligase mutant show a smear?
3. Why are human Okazaki fragments smaller than *E. coli*?
4. What does the FEN1 inhibitor result tell you about primer processing?

Expected Answers:

1. Primase mutant → fewer primers → longer fragments between primers
2. Ligase mutant → fragments not joined → all fragment sizes visible
3. Eukaryotes have more nucleosomes → shorter fragments needed to navigate chromatin
4. FEN1 is needed for flap removal → without it, fragments are slightly longer and contain flap structures

3. Replication Timing Analysis

Exercise: Interpreting Repli-seq Data

Simulated Data:

Chromosome 1 (partial, arbitrary region):

Position (Mb):	1.0	2.0	3.0	4.0	5.0	6.0	7.0
Repli-seq							
Early-S:	High	High	Low	Low	High	High	Low
Late-S:	Low	Low	High	High	Low	Low	High
Gene density:	High	High	Low	Low	High	High	Low

Analysis Questions:

1. What is the replication timing of the region around 1.0 Mb?
2. What is the replication timing of the region around 3.0 Mb?
3. What is the correlation between gene density and replication timing?
4. Predict the replication timing of a region with very high gene density.

Answers:

1. Early-S (replicates early in S phase)
2. Late-S (replicates late in S phase)
3. Positive correlation: high gene density → early replication
4. Very early-S (among the first regions to replicate)

4. Analyzing Telomere Length Data

Exercise: Telomere Length Measurement

Southern Blot (TRF) Data:

Sample: Genomic DNA digested with HinfI/RsaI + telomere probe

Lane 1: Newborn (healthy)
→ Broad smear centered at ~12 kb

Lane 2: Age 60 (healthy)
→ Broad smear centered at ~8 kb

Lane 3: Dyskeratosis congenita patient
→ Very short smear at ~3 kb

Lane 4: Cancer cell line (telomerase+)
→ Broad smear at ~10 kb

Lane 5: Cancer cell line (ALT+)
→ Very broad smear from ~2-25 kb

Questions:

1. How much telomere length is lost between birth and age 60?
2. What is the approximate annual rate of telomere shortening?
3. Why is the ALT cancer lane so broad compared to others?
4. Could the patient in Lane 3 have a mutation in TERT?

5. Sequence Analysis Exercises

Exercise: Strand Orientation Analysis

Task: Given a DNA sequence, determine:

1. Which strand is the template for leading strand synthesis
2. Where RNA primers would be placed on the lagging strand
3. The direction of fork movement

Example:

Given oriC sequence and surrounding region:

```
5'--...ATCGATCGATCGATCGATCG...[ORIC]...GCTAGCTAGCTAGCTAGCTA-3'
3'--...TAGCTAGCTAGCTAGCTAGC...[ORIC]...CGATCGATCGATCGATCGAT-5'
```

Left fork moves ← Right fork moves →

Exercise: Predicting Primer Binding

Task: Given a single-stranded template sequence, predict:

1. Where a primase would bind
2. The sequence of the RNA primer
3. The direction of extension

Example:

```
Template: 3'-GCTAGCTAGCTAGC-5'
          5'----->3' (synthesis direction)
```

If primase recognizes and binds at position 4-5:

RNA primer: 5'-rUrArG-3' (complementary to template)

Extension: 5'-rUrArG-dC-dG-dA-dC-dG-dA-dC-dG-3'

6. Mutation Analysis

Exercise: Predicting Consequences of Replication Gene Mutations

Scenario:

A patient presents with:

- Microsatellite instability (MSI) in colorectal cancer
- Family history of colorectal and endometrial cancer
- Tumor shows frameshift mutations in coding microsatellites

Question: Which genes are most likely mutated?

Analysis:

1. MSI indicates defective mismatch repair
2. Most common MMR gene mutations in Lynch syndrome: MLH1, MSH2, MSH6, PMS2

3. MLH1 and MSH2 account for ~70% of Lynch syndrome cases
4. Microsatellite instability diagnostic test confirms MMR deficiency

7. Comparative Genomics

Exercise: Comparing Replication Proteins Across Species

Task: Compare the sequences of DNA polymerase catalytic subunits across:

- E. coli (Pol III α , dnaE)
- Yeast (Pol ϵ catalytic, Pol2)
- Human (Pol ϵ catalytic, POLE)

Analysis:

1. Align sequences using BLAST
2. Identify conserved domains (polymerase, exonuclease)
3. Look for conserved catalytic residues
4. Note species-specific insertions/deletions

Quick Check

1. How would you use GC-skew analysis to predict the origin of replication in an uncharacterized bacterial genome?
2. If a new Okazaki fragment analysis shows fragments of ~500 nt in human cells, what might explain this unexpected result?
3. Design a bioinformatics workflow to identify potential replication origins in a newly sequenced bacterial genome.
4. A patient shows telomere length of 2 kb with premature aging features. What genetic tests would you recommend?

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Module 11: Clinical and Biotechnology Applications

Learning Objectives

1. Explain how replication mechanisms are targeted by drugs
2. Describe replication-related diseases and their molecular basis
3. Understand the role of replication in cancer biology
4. Connect replication to biotechnology applications
5. Understand replication in molecular diagnostics

1. Antibiotics Targeting Bacterial Replication

Concept

Because bacterial replication machinery differs from eukaryotic, it provides excellent targets for selective antibiotics.

Major Antibiotic Targets

Drug Class	Target	Mechanism	Clinical Use
Fluoroquinolones (Ciprofloxacin, Levofloxacin)	DNA Gyrase / Topo IV	Trap covalent complex → DSBs	Broad-spectrum
Rifampicin	RNA polymerase (β subunit)	Blocks transcription (indirectly affects replication initiation)	Tuberculosis
Novobiocin	DNA Gyrase (ATPase subunit)	Blocks ATP binding → prevents negative supercoiling	(Historical)
Nalidixic acid	DNA Gyrase	Similar to fluoroquinolones	(Historical)
Trimethoprim	Dihydrofolate reductase	Reduces dNTP pools → slows replication	UTI, respiratory
Hydroxyurea	Ribonucleotide reductase	Depletes dNTPs → stalls replication	Cancer, sickle cell

Fluoroquinolone Mechanism — Detailed

FLUOROQUINOLONE MECHANISM

Step 1: Gyrase binds DNA + fluoroquinolone

- Drug stabilizes covalent gyrase-DNA complex
- Normally transient DSB is stabilized

Step 2: Replication fork encounters stabilized complex

- Fork cannot pass
- Fork collapse
- Double-strand break

Step 3: DSBs accumulate

- SOS response activated
- If too many DSBs → cell death

Selectivity: Bacterial gyrase ≠ eukaryotic topoisomerases

- Selective toxicity

Clinical Application

- **Ciprofloxacin:** Used for urinary tract infections, respiratory infections, anthrax
- **Levofloxacin:** Community-acquired pneumonia
- **Moxifloxacin:** Tuberculosis (second-line)
- **Resistance:** Point mutations in *gyrA*/*gyrB*, plasmid-mediated quinolone resistance (PMQR)

2. Antiviral Drugs Targeting Replication

Concept

Many antiviral drugs target viral replication machinery, particularly polymerases.

Major Antiviral Targets

Drug	Target Virus	Target Protein	Mechanism
Acyclovir	HSV, VZV	Viral DNA polymerase	Nucleoside analog; chain terminator
Ganciclovir	CMV	Viral DNA polymerase	Nucleoside analog
Zidovudine (AZT)	HIV	Reverse transcriptase	Nucleoside analog; chain terminator
Tenofovir	HIV, HBV	Reverse transcriptase	Nucleotide analog
Sofosbuvir	HCV	RNA-dependent RNA polymerase	Nucleotide analog
Remdesivir	SARS-CoV-2	RdRp	Nucleotide analog
Molnupiravir	SARS-CoV-2	RdRp	Mutagenic nucleoside analog

Nucleoside Analog Mechanism

NUCLEOSIDE ANALOG MECHANISM

Step 1: Drug enters cell

→ Prodrug (acyclovir) or active form

Step 2: Phosphorylation by cellular/viral kinases

→ Acyclovir → ACV-MP → ACV-DP → ACV-TP (active)

Step 3: Incorporated by viral polymerase

→ ACV-TP competes with dGTP

→ Incorporated into growing DNA

Step 4: Chain termination

→ ACV lacks 3'-OH

→ No further nucleotides can be added

→ DNA synthesis terminates

Selectivity: Viral polymerase incorporates analog more efficiently than cellular polymerases

3. Cancer Biology and Replication

Concept

Cancer cells have abnormal replication: increased origin firing, replication stress, checkpoint defects, and telomere maintenance.

Replication in Cancer

Feature	Normal Cells	Cancer Cells
Origin firing	Regulated, once per cycle	Excessive, re-replication possible
Checkpoint	Intact ATR/CHK1	Often defective
Telomerase	Inactive (most somatic)	Active (~85-90% of cancers)
Replication stress	Low	High (oncogene-induced)
DNA repair	Intact	Often defective
DNp53	Wild type	Often mutated

Cancer Therapies Targeting Replication

Therapy	Target	Mechanism
PARP inhibitors (Olaparib)	PARP1/2	Synthetic lethality in BRCA-mutant cancers
ATR inhibitors (Ceralasertib)	ATR kinase	Sensitize to replication stress
CHK1 inhibitors (Prexasertib)	CHK1 kinase	Abrogate S/G2 checkpoint
TERT inhibitors (Imetelstat)	Telomerase	Inhibit telomere maintenance

Therapy	Target	Mechanism
Topoisomerase inhibitors	Topo I/II	Trap complexes → DSBs
Antimetabolites (5-FU, Gemcitabine)	Nucleotide metabolism	Deplete dNTPs
Hydroxyurea	Ribonucleotide reductase	Deplete dNTPs

PARP Inhibitor — Synthetic Lethality

SYNTHETIC LETHALITY: PARP INHIBITION IN BRCA-MUTANT CANCERS

Normal cells:

BRCA1/2 functional → HR repair works
 PARP inhibited → BER impaired → SSBs accumulate
 → SSBs converted to DSBs during replication
 → HR repair fixes DSBs → cell survives

BRCA-mutant cancer cells:

BRCA1/2 defective → HR repair deficient
 PARP inhibited → BER impaired → SSBs accumulate
 → SSBs converted to DSBs during replication
 → HR repair cannot fix DSBs → CELL DEATH

Result: Selective killing of BRCA-mutant cancer cells

4. Genetic Diseases of Replication

Disease	Gene(s)	Replication Defect
Lynch syndrome (HNPCC)	MLH1, MSH2, MSH6, PMS2	Defective mismatch repair
Dyskeratosis congenita	TERT, TERC, DKC1	Defective telomere maintenance
Fanconi anemia	FANCA, BRCA2, etc.	Defective interstrand crosslink repair
Bloom syndrome	BLM (RecQ helicase)	Defective replication fork processing
Werner syndrome	WRN (RecQ helicase)	Premature aging, defective replication
MRE11 ATLD	MRE11	Defective DSB repair
Nijmegen breakage syndrome	NBS1	Defective checkpoint response
ATR-Seckel syndrome	ATR	Defective replication checkpoint

5. Biotechnology Applications

DNA Sequencing

Technology	Replication Principle Used
Sanger sequencing	Chain termination (dideoxynucleotides)
Illumina sequencing	Bridge amplification, reversible terminators

Technology	Replication Principle Used
PacBio SMRT	Single-molecule polymerase observation
Oxford Nanopore	Direct DNA translocation

PCR (Polymerase Chain Reaction)

PCR is a laboratory technique inspired by DNA replication:

- Uses thermostable DNA polymerase (Taq, Pfu)
- Uses specific primers (like in vivo primers)
- Requires dNTPs (like in vivo substrates)
- Does NOT have proofreading (Taq) or has proofreading (Pfu)
- **Key difference from cellular replication:** No helicase (heat denatures DNA), no sliding clamp, no Okazaki fragments

Molecular Cloning

Replication components used in cloning:

- **Origin of replication** (plasmid): Allows plasmid to replicate in host
- **Selectable markers:** Antibiotic resistance
- **Restriction enzymes:** Cut DNA at specific sites
- **DNA ligase:** Join fragments

Synthetic Biology

- **Synthetic origins:** Design custom replication origins
- **Minimal genomes:** Understanding essential replication genes
- **Xenobiology:** Unnatural nucleotides that can be replicated

6. Molecular Diagnostics

Applications

Diagnostic	Replication Principle
BRCA1/2 testing	Homologous recombination deficiency
MSI testing	Mismatch repair deficiency
TERT promoter testing	Telomerase activation
CTC enumeration	Replication markers (Ki67, PCNA)
Liquid biopsy	ctDNA analysis
Microsatellite analysis	Replication slippage detection

Microsatellite Instability (MSI) Testing

MSI TESTING IN CANCER

Principle: Microsatellites (repeats) are prone to slippage during replication
MMR corrects slippage errors
MMR deficiency → MSI

Method:

1. PCR amplify microsatellite markers (BAT25, BAT26, D2S123, etc.)
2. Compare tumor vs. normal
3. MSI-High: $\geq 2/5$ markers unstable
4. MSI-Low: 1/5 marker unstable
5. MSS (microsatellite stable): 0/5 markers unstable

Clinical significance:

- MSI-High → Lynch syndrome likely
- MSI-High → better prognosis in colorectal cancer
- MSI-High → responsive to immune checkpoint inhibitors

7. Telomere-Based Diagnostics

Telomere Length as Biomarker

Application	Use
Aging research	Telomere length correlates with biological age
Cancer prognosis	Short telomeres may predict outcome
Bone marrow failure	Diagnostic for dyskeratosis congenita
Cardiovascular disease	Short telomeres associated with risk
Psychological stress	Chronic stress associated with shorter telomeres

Quick Check

1. Why are fluoroquinolones selectively toxic to bacteria but not human cells?
2. Explain the principle of synthetic lethality and how PARP inhibitors exploit this in BRCA-mutant cancers.
3. A patient presents with both colorectal and endometrial cancer. Which replication-associated genes should be tested?
4. How does PCR relate to cellular DNA replication? What are the key differences?
5. Explain why MSI testing is useful for both diagnosing Lynch syndrome and predicting immunotherapy response.

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Module 12: Advanced Topics in DNA Replication

Learning Objectives

1. Describe replication fork dynamics in detail
2. Explain fork stalling, reversal, and restart mechanisms
3. Describe replication-transcription conflicts and R-loops
4. Explain the ATR-CHK1 pathway in detail
5. Understand translesion synthesis and template switching
6. Explain common fragile sites and their role in cancer
7. Describe alternative lengthening of telomeres (ALT)
8. Understand genome instability mechanisms

1. Replication Fork Dynamics

Concept

The replication fork is not a static structure — it undergoes dynamic changes in composition, speed, and stability in response to cellular conditions.

Fork Speed Regulation

Condition	Effect on Fork Speed
Normal conditions	50 nt/s (human), 1000 nt/s (E. coli)
Low dNTPs	Slowed
Transcription-active regions	Slowed (conflicts)
Chromatin-dense regions	Slowed
DNA damage	Stalled
Replication stress	Stalled or reversed

Fork Protection Complex

The fork protection complex (FPC) maintains fork stability:

- **Timeless (Timestop):** Fork-associated; recruits ATR
- **Claspin:** Mediator; bridges MCM-ATRIP
- **Mrc1/Claspin:** Stabilizes fork; promotes restart

2. Fork Stalling and Collapse

Concept

When a replication fork encounters an obstacle (DNA damage, secondary structure, transcription complex), it can:

1. **Pause** (temporary, recoverable)
2. **Stall** (longer pause, may require checkpoint activation)
3. **Collapse** (fork breakage → double-strand break → genome instability)

Fork Collapse

FORK COLLAPSE

Intact stalled fork:

```

3'=====→5' (lagging template)
←===== (new lagging)

5'=====→3' (leading template)
←===== (new leading, stopped)

```

Collapsed fork (DSB):

```

3'=====→5'
←=====
                X DSB

5'=====→3'

```

- Requires HR for repair
- Risk of chromosome rearrangement

Fork Reversal (Regression)

FORK REVERSAL

Reversed fork ("chicken foot"):

```

3'=====→5' (lagging template)
←===== (new lagging, annealed to leading)
      |
5'=====→3' (leading template)
←===== (new leading, annealed to lagging)

```

- New strands anneal to each other
- Creates a 4-way junction
- Allows template switching or lesion bypass

Key enzymes for fork reversal:

- SMARCAL1 (SWI/SNF family translocase)
- ZRANB3 (Holliday junction resolvase-like)
- HLTF (SWI/SNF family)
- RAD51 (facilitates regression)

3. Replication-Transcription Conflicts

Concept

DNA replication and transcription use the same DNA template. When a replication fork encounters an active transcription complex, they can collide, causing fork stalling, DNA damage, and genome instability.

Types of Conflicts

REPLICATION-TRANSCRIPTION CONFLICTS

HEAD-ON CONFLICT (most dangerous):

← Fork → RNA Pol



Fork and RNA polymerase move toward each other

- Direct collision
- Fork collapse likely
- DSB formation

CO-TRANSCRIPTIONAL (less dangerous):

→ Fork → RNA Pol



Fork catches up to RNA polymerase from behind

- Fork may push RNA polymerase off
- Less likely to cause breakage

R-Loops

R-loops are three-stranded structures formed when RNA hybridizes to the template DNA strand, displacing the non-template strand.

R-LOOP STRUCTURE

Normal DNA:

5'—=====3' (template)
3'—=====5' (non-template)

R-loop:

5'====RNA====3' (template hybridized with RNA)
3'====5' (displaced non-template → ssDNA)



Features:

- RNA:DNA hybrid: ~100-300 bp typically
- Displaced ssDNA: prone to damage
- Can block replication forks
- Common at highly transcribed genes
- Regulated by RNase H1/H2 (degrade RNA in R-loops)

R-Loop Regulation

Factor	Function
RNase H1/H2	Degrade RNA in R-loops
Senataxin	RNA/DNA helicase; resolves R-loops
SETX (Senataxin)	Helicase activity
Topoisomerase I	Relieves torsional stress that promotes R-loops
ASF1	Histone chaperone; prevents R-loop formation
BRCA1	Facilitates R-loop resolution

4. Translesion Synthesis (TLS)

Concept

Translesion synthesis is a DNA damage tolerance mechanism where specialized low-fidelity polymerases bypass replication-blocking lesions, allowing fork progression at the cost of increased mutagenesis.

TLS Polymerases in Humans

Polymerase	Specialization	Error Rate
Pol η	Bypasses UV pyrimidine dimers (T-T CPDs)	Low for CPDs
Pol ι	Bypasses UV lesions, some oxidized bases	Moderate
Pol κ	Bypasses bulky adducts	Moderate
Rev1	dCMP transferase; scaffolding	High
Pol ζ	Extension from mismatched primer termini	High

TLS Mechanism

TRANSLATION SYNTHESIS

Normal replication:

- Pol ϵ/δ encounters lesion
- Polymerase stalls (can't bypass)

TLS switching:

1. PCNA is mono-ubiquitylated (K164)
2. Ubiquitylated PCNA recruits TLS polymerase
3. TLS polymerase replaces Pol ϵ/δ at primer terminus
4. TLS polymerase incorporates 1-2 nucleotides opposite lesion
5. TLS polymerase dissociates
6. Pol ϵ/δ resumes replication (with PCNA de-ubiquitylated)

Result: Lesion bypassed (at cost of potential mutation)
Fork progresses (genome integrity maintained)

Template Switching vs. TLS

Feature	Template Switching	Translesion Synthesis
Fidelity	Error-free	Error-prone
Template	Sister chromatid	Lesion-containing strand
Polymerase	Pol ϵ/δ (same as replication)	Specialized TLS polymerases
PCNA modification	Poly-ubiquitylation (K63)	Mono-ubiquitylation (K164)
Outcome	No mutation	Potential mutation

5. Homologous Recombination During Replication

Concept

Homologous recombination (HR) is essential for repairing replication-associated DNA breaks and restarting collapsed replication forks.

HR at Collapsed Forks

HR AT COLLAPSED FORK

- Step 1: Collapsed fork generates DSB
→ 5' end resection by MRN complex + CtIP
- Step 2: 3' ssDNA overhang generated
→ RPA binds
- Step 3: RAD51 replaces RPA (with BRCA2 assistance)
→ RAD51 nucleoprotein filament forms
- Step 4: Strand invasion of sister chromatid
→ D-loop formation
- Step 5: DNA synthesis using sister chromatid as template
- Step 6: Resolution/dissolution of recombination intermediates
- Step 7: Fork restart

BRCA1/2 in Replication

Protein	Function
BRCA1	Facilitates end resection; recruits repair factors
BRCA2	Loads RAD51 onto ssDNA
PALB2	Bridges BRCA1 and BRCA2

BRCA1/2 mutations → defective HR → genome instability → cancer predisposition (breast, ovarian)

6. Common Fragile Sites (CFS)

Concept

Common fragile sites are specific genomic regions prone to breakage under replication stress. They are large, gene-rich regions that are difficult to replicate.

Why They Break

1. **Large genes:** Some CFS genes are >2 Mb long
2. **Late replication:** Despite having early origins, some regions replicate late
3. **Transcription-replication conflicts:** Highly transcribed genes
4. **Difficult-to-replicate sequences:** AT-rich, prone to secondary structures
5. **Origin paucity:** Few origins in large regions

CFS in Cancer

CFS	Gene	Cancer Association
FRA3B	FHIT	Lung, breast, gastric
FRA16D	WWOX	Breast, ovarian, leukemia
FRA6E	PARK2	Colorectal, ovarian
FRA7H	Various	Multiple cancers

CFS Expression

COMMON FRAGILE SITE EXPRESSION

Normal conditions:

- CFS are intact (rarely break)
- Adequate time for replication

Mild replication stress (low aphidicolin):

- CFS show gaps/breaks in metaphase chromosomes
- Large genes haven't finished replicating
- Indicates regions prone to breakage

Severe replication stress:

- CFS break frequently
- Chromosome rearrangements at CFS
- → Cancer development

7. Oncogene-Induced Replication Stress — Detailed

Mechanisms of Oncogene-Induced Stress

Oncogene	Mechanism of Stress
Ras	Aberrant origin firing, increased RNR, S-phase entry

Oncogene	Mechanism of Stress
Myc	Aberrant origin firing, increased transcription, replication-transcription conflicts
Cyclin E	Premature S-phase entry, incomplete licensing
E2F	Premature origin firing
Cdc25A	Premature CDK activation, origin firing
BRCA1 loss	Defective fork protection

Genome Instability Cascade

ONCOGENE ACTIVATION → REPLICATION STRESS → GENOME INSTABILITY

```

Oncogene (e.g., Ras) activated
↓
Excessive origin firing
↓
Dormant origins activated (should be silent)
↓
Fork collisions (forks converge inappropriately)
↓
Reversed/collapsed forks
↓
Double-strand breaks
↓
If checkpoint intact → cell death or senescence
↓ (p53 loss)
Checkpoint adaptation
↓
Cell survives with damaged genome
↓
Chromosomal rearrangements accumulate
↓
Cancer progression

```

8. Genome Instability Mechanisms

Types of Genome Instability

Type	Mechanism	Consequence
Point mutations	Replication errors, TLS	Amino acid changes
Insertions/Deletions	Replication slippage	Frameshifts, repeat changes
Copy number variations	Re-replication, NHR	Gene amplification/deletion
Translocations	NHR between different chromosomes	Oncogene activation
Chromosomal instability	Defective mitosis, breakage	Aneuploidy
Microsatellite instability	MMR deficiency	Frameshift mutations

Repeat Instability

Repeat Type	Disease	Mechanism
Trinucleotide (CAG)	Huntington disease	Replication slippage → expansion
Trinucleotide (CGG)	Fragile X syndrome	Replication slippage → expansion
Trinucleotide (CTG)	Myotonic dystrophy	Replication slippage → expansion
Trinucleotide (GAA)	Friedreich ataxia	Replication slippage → expansion

9. Alternative Lengthening of Telomeres (ALT)

Concept

ALT is a telomerase-independent mechanism for telomere maintenance that uses homologous recombination between telomeric sequences.

ALT Mechanism

ALT MECHANISM

- Step 1: Telomeric DSB or stall
→ Break in telomeric repeat region
- Step 2: Resection
→ 3' overhang generated
- Step 3: Strand invasion
→ 3' overhang invades another telomere (sister or homolog)
→ Uses other telomere as template
- Step 4: DNA synthesis
→ Extends telomere using homologous telomere as template
- Step 5: Resolution
→ Recombination intermediates resolved
→ Extended telomere

Features of ALT:

- Highly heterogeneous telomere lengths (2-50 kb)
- ALT-associated PML bodies (APBs)
- C-circles (partially single-stranded circular telomeric DNA)
- Homologous recombination dependent

ALT Markers

Marker	Detection Method
APBs	FISH for telomeric DNA + PML immunofluorescence
C-circles	C-circle assay (rolling circle amplification)
Telomere length heterogeneity	TRF Southern blot

Marker	Detection Method
ALT-associated recombination	Telomere-specific FISH

10. Replication Licensing Defects

Concept

Defective licensing control can lead to:

- **Re-replication:** Origins fire more than once → gene amplification
- **Under-replication:** Not enough origins fire → incomplete replication

Re-replication Consequences

RE-REPLICATION

Normal: One round per cell cycle

Origin fires → complete replication → done

Re-replication (licensing defect):

Origin fires → partial replication → re-licenses → fires again

↓

Gene amplification

↓

Oncogene activation

↓

Cancer

Genes That Prevent Re-replication

Gene	Function	Cancer Association
Geminin	Inhibits Cdt1	Lost in some cancers
CDK (multiple)	Phosphorylate replication factors	Dysregulated in cancer
APC/C	Degrades geminin, CDC20	Mutated in some cancers
Cdt1	MCM loader (must be inhibited in S)	Overexpressed in some cancers

Quick Check

1. Explain why replication-transcription conflicts are particularly problematic at highly transcribed genes.
2. Describe how translesion synthesis contributes to both cell survival and mutagenesis.
3. How does fork reversal differ from fork collapse? Which is more dangerous?
4. Explain the relationship between BRCA1/2 mutations, homologous recombination, and PARP inhibitor sensitivity.
5. What molecular features distinguish ALT-positive from telomerase-positive cancers?

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43. **PDB (Protein Data Bank):** <https://www.rcsb.org/>
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44. **REBASE (Restriction Enzyme Database):** <http://rebase.neb.com/>
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45. **Oncogene Census:** <https://cancer.sanger.ac.uk/cosmic/census>
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Online Educational Resources

46. **MIT OpenCourseWare:** Biology (7.013, 7.014, 7.016)
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47. **Khan Academy:** DNA replication
 - Beginner-friendly video explanations
48. **HHMI BioInteractive:** DNA Replication
 - Interactive animations and resources
49. **DNA Learning Center (CSHL):** <https://www.dnalc.org/>
 - DNA science resources
50. **NCBI Bookshelf:** Molecular Biology of the Cell (Alberts et al.)
 - Free online access to textbook chapters

Image and Diagram Sources

51. All diagrams in this course are original or adapted from:
 - Alberts et al., *Molecular Biology of the Cell* (with appropriate attribution)
 - Lodish et al., *Molecular Cell Biology*
 - Watson et al., *Molecular Biology of the Gene*
52. Scientific illustrations created for this course are based on published structural and mechanistic data from peer-reviewed literature.

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